



Modelling Financial Markets

Seminar 18: Systemic Risk and Networks — Hands-on



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Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BIC:** Bayesian Information Criterion
- ▣ **BNP:** BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ **CDS:** Credit Default Swap
- ▣ **COVID-19:** COronaVirus Disease 2019
- ▣ **DBK:** Deutsche Bank AG (Xetra ticker)
- ▣ **DCC:** Dynamic Conditional Correlation
- ▣ **ES:** Expected Shortfall
- ▣ **FEVD:** Forecast Error Variance Decomposition
- ▣ **FRM:** Financial Risk Meter
- ▣ **FWER:** Family-Wise Error Rate
- ▣ **GACV:** Generalized Approximate Cross-Validation
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GJR:** Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ **HAC:** Heteroskedasticity and Autocorrelation Consistent (standard errors)

Acronyms used in this material (2/2)

- ▣ **HSBC**: HSBC Holdings plc (originally the Hongkong and Shanghai Banking Corporation)
- ▣ **ING**: ING Groep (Internationale Nederlanden Groep) — *the Dutch banking group ING*
- ▣ **JPM**: JPMorgan Chase & Co. (NYSE ticker)
- ▣ **LRMES**: Long-Run Marginal Expected Shortfall
- ▣ **MES**: Marginal Expected Shortfall
- ▣ **OLS**: Ordinary Least Squares
- ▣ **QFVAR**: Quantile Factor Vector AutoRegression
- ▣ **QML**: Quasi-Maximum Likelihood
- ▣ **RTD**: Relative Tail Dependence
- ▣ **SG**: Société Générale — *the French banking group Société Générale*
- ▣ **SPX**: Standard and Poor's 500 index (column name in the notebook)
- ▣ **SRISK**: Systemic RISK (capital shortfall measure)
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression
- ▣ **VIX**: CBOE Volatility Index

Today's Questions and Expected Results

- Questions of the seminar
 - ▶ which banks are systemic?
 - ▶ how sure can we be?
- Route
 - ▶ First the primer “Prerequisites for Today” (the lecture comes after the seminar); then A1, A3, A5 on paper: CoVaR, MES and the generalized variance decomposition
 - ▶ B1: test whether the US banks differ in ΔCoVaR
 - ▶ B5, B9: how robust and how precise is the connectedness network?
 - ▶ C2: find the errors in an AI answer; B2 at home; B3 (LRMES by GARCH–DCC simulation) after the lecture
- What you produce: the A answers, one table and one chart per B exercise, the corrected C2 numbers and error log, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); each task names its notebook section
- Background: the primer below; [Chapter 18](#) follows after the seminar; textbook [Franke, Härdle & Hafner \(2019\)](#)

Exercise Map

Exercise	Question	Type	Model
A1	ΔCoVaR under the Normal distribution	Solved	–
A3	why MES adds up to the portfolio ES	Solved	–
A5	the generalized variance decomposition	Solved	–
B1	do US banks differ in ΔCoVaR 5%?	Solved	–
B5, B9	robustness and intervals of connectedness	Solved	–
C2	what is wrong in an AI answer?	Proposed	A1, A3
B2	European banks: which is most systemic?	Proposed	B1
Extension packs	A2, A4, A6–A8, B3, B4, B6–B8, B10, C1, C3 (optional, after the lecture)	Solved and Proposed (marked on each task)	named on each task

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: estimation and inference on real data, each ending with an interpretation question; Part C: critique of an AI answer and open questions

Data Used

Series	Assets	Price used	Used in
US banks	JPMorgan, Bank of America, Citigroup, Goldman Sachs, Morgan Stanley, Wells Fargo	adjusted close; daily high and low	B1, B3, B5, B9
European banks	Deutsche Bank, BNP Paribas, Santander, ING, HSBC	adjusted close; daily high and low	B2, B5, B9
Romanian banks	Banca Transilvania, BRD-Groupe SG (days without trades removed)	adjusted close; daily high and low	B5, B9
Market indices	S&P 500, Euro Stoxx 50, BET, VIX	close	B3, packs

- Daily prices of the course data, one series per asset (field used: adjusted close for the banks, close for the indices; daily high and low for volatility); sample 5 January 2010 – 18 September 2026
- Returns R : prices aligned first on the 3960 days common to all 17 series, then daily log returns in %; volatilities V : log Parkinson volatility (from the daily high–low range) of the 13 banks

Setup: Run and Verify

```
# notebook section "Data" builds
# R (returns, %) and V (log volatility)
print(R.shape, V.shape)
print(R.index[0].date(),
      R.index[-1].date())
print(R[['JPM', 'DBK', 'TLV',
        'SPX']].head(3).round(2))
```

- Expected output: (3960, 17) (3966, 13); first and last day 2010-01-05, 2026-09-18; V needs only the 13 banks, not the indices, so it has more common days (3966)

date	JPM	DBK	TLV	SPX
2010-01-05	1.92	1.60	2.73	0.31
2010-01-06	0.55	-1.05	3.09	0.05
2010-01-07	1.96	0.45	0.00	0.40

- If your counts or first values differ, check the symbols, that you use the dividend-adjusted price, and the alignment on common days

Prerequisites for Today (1/23): Systemic Risk, Quantiles, VaR and ES

- The seminar comes **before** the lecture
 - ▶ These frames give only what today's tasks need; the rest in [Chapter 18](#), after the seminar
- Two questions about a bank
 - ▶ **Exposure**: how much does the bank lose when the system is in trouble? (MES, LRMES, SRISK)
 - ▶ **Contribution**: how much worse is the system when the bank is in trouble? (ΔCoVaR)
- The α -quantile $q_\alpha(X)$ of a return X : the value it falls below with probability α

$$P(X \leq q_\alpha(X)) = \alpha, \quad \text{VaR}_\alpha = -q_\alpha(X), \quad \text{ES}_\alpha = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$$

- ▶ α : tail probability (1% or 5%); VaR: the loss exceeded on a fraction α of days; ES: the average loss on those days, so $\text{ES}_\alpha \geq \text{VaR}_\alpha$
- ▶ $\mathbb{E}[\cdot \mid A]$: mean conditional on the event A ; VaR, ES, CoVaR, MES are reported as positive losses (%)

Prerequisites for Today (2/23): CoVaR and ΔCoVaR : Definitions

- CoVaR_α : the VaR of the system X^{sys} on the bank's distress day [Adrian & Brunnermeier \(2016\)](#)

$$P(X^{\text{sys}} \leq -\text{CoVaR}_\alpha \mid X_i = q_\alpha(X_i)) = \alpha$$

- ▶ X_i : return of bank i ; X^{sys} : return of the system (here the other banks of the region); $X_i = q_\alpha(X_i)$: the bank is at its own VaR
- The same quantity in the bank's median state, $X_i = q_{50\%}(X_i)$: $\text{CoVaR}_\alpha^{50\%}$
 - ▶ $q_{50\%}(X_i)$: the median return, a normal day for the bank
- The contribution of the bank: the difference

$$\Delta\text{CoVaR}_\alpha = \text{CoVaR}_\alpha - \text{CoVaR}_\alpha^{50\%}$$

- ▶ the extra tail loss of the system when the bank moves from a normal day to distress; > 0 when bank and system move together
- ▶ own risk (VaR of the bank) and contribution (ΔCoVaR) are different quantities (A1, B2)

Prerequisites for Today (3/23): Estimating CoVaR by Quantile Regression

- Quantile regression: a line for the α -quantile of X^{sys} , not for its mean

$$q_{\alpha}(X^{\text{sys}} | X_i) = a_{\alpha} + b_{\alpha} X_i, \quad (\hat{a}_{\alpha}, \hat{b}_{\alpha}) = \arg \min_{a, b} \sum_{t=1}^n \rho_{\alpha}(X_t^{\text{sys}} - a - b X_{i,t})$$

- b_{α} : by how much the α -quantile of the system moves per point of X_i ; n : number of days; hat: estimated value

- The **check function** replaces the squared residual of OLS

$$\rho_{\alpha}(u) = u(\alpha - 1\{u < 0\})$$

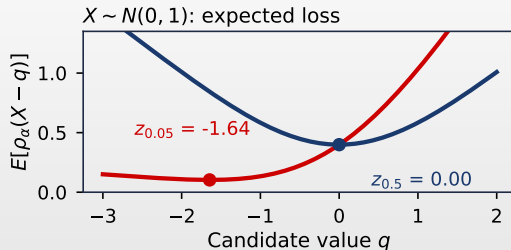
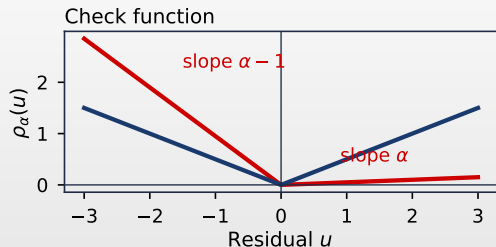
- $1\{\cdot\}$: 1 if the condition holds, else 0; negative residuals weigh $1 - \alpha$, positive ones α , so a share α of the points lies below the line

- Plug in the empirical quantiles $\hat{q}_{\alpha}(X_i)$, $\hat{q}_{50\%}(X_i)$ of the bank

$$\widehat{\text{CoVaR}}_{\alpha} = -(\hat{a}_{\alpha} + \hat{b}_{\alpha} \hat{q}_{\alpha}(X_i)), \quad \Delta \widehat{\text{CoVaR}}_{\alpha} = \hat{b}_{\alpha} (\hat{q}_{50\%}(X_i) - \hat{q}_{\alpha}(X_i))$$

- the slope times the distance from the median day to the distress day; $\hat{a}_{\alpha}$ cancels

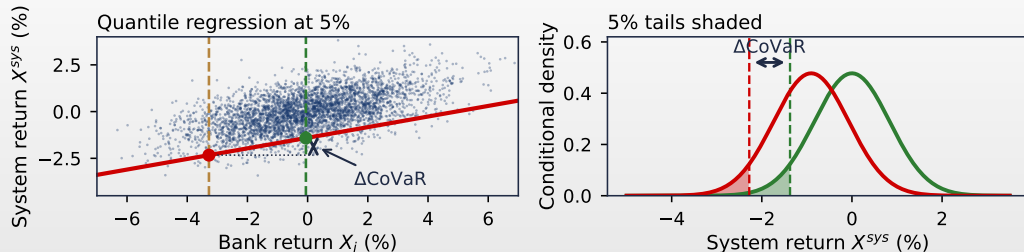
Prerequisites for Today (4/23): The Check Function



— $\alpha = 0.05$ — $\alpha = 0.5$

- Left: $\rho_\alpha(u)$ is a tilted absolute value; at $\alpha = 0.5$ it is $|u|/2$ (median regression)
- Right: for $X \sim N(0, 1)$, $\mathbb{E}[\rho_\alpha(X - q)]$ is smallest at the α -quantile: -1.64 at 5%, 0 at 50%
 - minimising ρ_α estimates a quantile, as minimising squares estimates a mean

Prerequisites for Today (5/23): ΔCoVaR on Simulated Data



— 5% line $\hat{a} + \hat{b}X_i$
 - - - distress $\hat{q}_{5\%}(X_i)$
 - - - median $\hat{q}_{50\%}(X_i)$
 — $X^{sys} \mid \text{median}$
 — $X^{sys} \mid \text{distress}$

- Simulated bivariate Normal returns: $\sigma_i = 2\%$, $\sigma_{sys} = 1\%$, $\rho = 0.55$, $n = 4\,000$ days
 - ▶ left: slope $\hat{b}_{5\%} = 0.28$; from the median day to the distress day ($\hat{q}_{5\%} = -3.28\%$) the line falls by $\Delta\text{CoVaR} = 0.91\%$
 - ▶ right: the 5% quantiles (dashed) of the two conditional densities of X^{sys} are ΔCoVaR apart
- Normal closed form [Adrian & Brunnermeier \(2016\)](#): $\Delta\text{CoVaR}_\alpha = \rho \sigma_{sys} |z_\alpha|$, $z_\alpha = \Phi^{-1}(\alpha)$ (Φ : standard Normal distribution function); here $0.55 \cdot 1\% \cdot 1.645 = 0.90\%$

Prerequisites for Today (6/23): A Worked ΔCoVaR Example

- ▣ Estimates: $\hat{b}_{5\%} = 0.5$, $\hat{q}_{50\%} = 0.05\%$, $\hat{q}_{5\%} = -2.95\%$
 - ▶ $\Delta\text{CoVaR}_{5\%} = 0.5 \cdot (0.05 + 2.95) = 1.5\%$ of extra system loss
- ▣ Normal case: $\rho = 0.4$, $\sigma_{\text{sys}} = 1\%$, $\alpha = 5\%$
 - ▶ $\Delta\text{CoVaR}_{5\%} = 0.4 \cdot 1\% \cdot 1.645 = 0.66\%$
 - ▶ it does not involve σ_i : a riskier bank with the same ρ has the same contribution (A1)
- ▣ In B1, B2: the system of bank i excludes bank i
 - ▶ $X_t^{\text{sys}} = \frac{1}{N-1} \sum_{j \neq i} X_{j,t}$, the equal-weighted return of the other $N - 1$ banks of the region

Prerequisites for Today (7/23): Exposure: MES and ES

- **Marginal expected shortfall** Acharya et al. (2017): the bank's average loss on the market's worst α days

$$\text{MES}_{i,\alpha} = -\mathbb{E}[X_i \mid X_m \leq q_\alpha(X_m)]$$

- ▶ X_m : return of the market (or of the portfolio); the condition is on the **market's** tail, not the bank's

- Additivity (A3): with portfolio weights w_i , the weighted MES add up to the ES of the portfolio

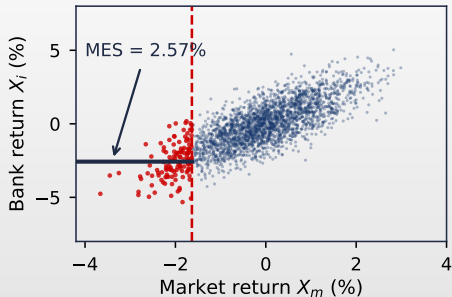
$$\sum_i w_i \text{MES}_{i,\alpha} = \text{ES}_\alpha(X_p), \quad X_p = \sum_i w_i X_i$$

- Normal case: MES is beta times the ES of the market

$$\text{MES}_{i,\alpha} = \beta_i \text{ES}_\alpha(X_m), \quad \beta_i = \frac{\rho \sigma_i}{\sigma_m}, \quad \text{ES}_\alpha(X_m) = \frac{\sigma_m \varphi(z_\alpha)}{\alpha}$$

- ▶ σ_i, σ_m : standard deviations of bank and market; ρ : their correlation; φ : standard Normal density
- ▶ e.g. $\beta = 1.2, \sigma_m = 1\%, \alpha = 5\%$: $\text{ES} = 2.06\%, \text{MES} = 2.47\%$

Prerequisites for Today (8/23): MES on a Scatter Plot



- other days
- market's worst 5%
- $q_5\%(X_m) = -1.64\%$
- mean of X_i there = $-\text{MES}$

- Simulated days: $X_i = 1.2 X_m + e_i$, $\sigma_m = 1\%$
 - ▶ e_i : the bank's own shock, independent of the market, standard deviation 1%
- Red: the 125 days with $X_m \leq q_5\%(X_m)$
 - ▶ $\text{MES} = \text{minus the mean of } X_i \text{ on these days (horizontal line)}$
 - ▶ estimate 2.57%, against the Normal value $\beta \text{ ES} = 2.47\%$: the difference is sampling error from 125 days
- A bank with $\rho = 0$ would have $\text{MES} \approx 0$, whatever its own volatility
 - ▶ MES measures co-movement in the tail, not own risk

Prerequisites for Today (9/23): Exposure in a Crisis: LRMES

- **Long-run MES (LRMES)** [Brownlees & Engle \(2017\)](#): the expected fall of the bank's equity if the market crashes

$$\text{LRMES}_i = -\mathbb{E}[R_{i,t+1:t+h} \mid R_{m,t+1:t+h} < C]$$

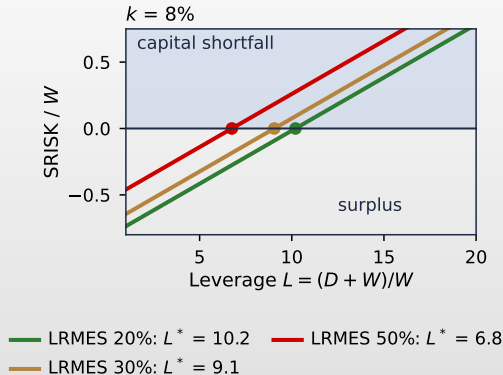
- ▶ $R_{t+1:t+h}$: arithmetic return over the next h days; C : crisis threshold of the market
- ▶ scenarios: $h = 22$ days, $C = -10\%$ (one month); $h = 126$ days, $C = -40\%$ (six months)

- By simulation (B3): GARCH volatilities and DCC correlation, many paths, average over the crisis paths
- By a shortcut from daily data [Acharya, Engle & Richardson \(2012\)](#)

$$\text{LRMES}_i \approx 1 - \exp(-18 \text{MES}_i^{2\%}), \quad \text{MES}_i^{2\%} = -\mathbb{E}[X_i \mid X_m < -2\%]$$

- ▶ X as fractions; $\text{MES}^{2\%}$: average daily bank loss on days when the market falls by more than 2%; the 18 is derived in A4

Prerequisites for Today (10/23): Exposure in a Crisis: SRISK



- **SRISK** Brownlees & Engle (2017): capital the bank would lack in the crisis

$$\text{SRISK} = k[D + (1 - \text{LRMES})W] - (1 - \text{LRMES})W$$

- ▶ k : prudential capital ratio; D : book debt; W : market equity; required capital minus remaining equity
- ▶ equivalently $kD - (1 - k)(1 - \text{LRMES})W$

- Zero at the break-even leverage L^*

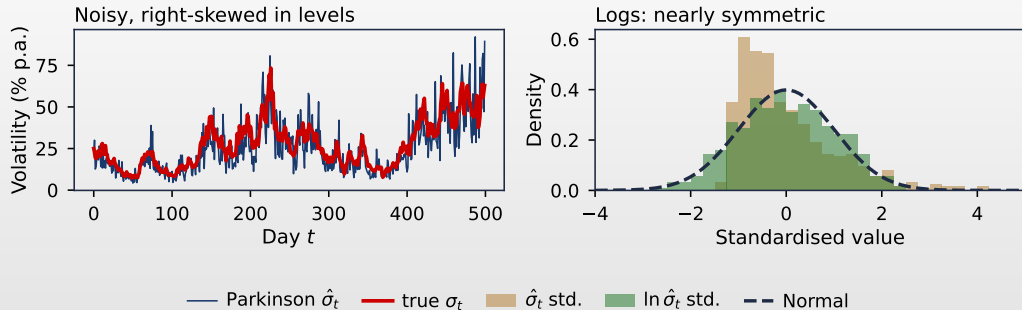
$$L^* = 1 + \frac{(1 - k)(1 - \text{LRMES})}{k}, \quad L = \frac{D + W}{W}$$

- ▶ $L > L^*$: shortfall (chart: above zero); a larger LRMES lowers L^*

- Example: $W = 50$, $D = 500$, $k = 8\%$, LRMES = 30%

- ▶ $\text{SRISK} = 40 - 0.92 \cdot 0.70 \cdot 50 = 7.8$;
 $L = 11 > L^* = 9.05$

Prerequisites for Today (11/23): Parkinson Volatility from the Daily Range



□ Parkinson estimate $\hat{\sigma}_t = \sqrt{252 (\ln H_t - \ln L_t)^2 / (4 \ln 2)}$; H_t, L_t : daily high and low

- ▶ $4 \ln 2$ makes $\hat{\sigma}_t^2$ unbiased for a Brownian price without drift; 252: annualisation
- ▶ left: one-day estimates (blue) are noisy around the simulated true volatility (red)
- ▶ right: skewness 1.3 in levels, 0.0 in logs; the VAR is therefore fitted to $\ln \hat{\sigma}_t$

Prerequisites for Today (12/23): The Vector Autoregression (VAR)

- VAR(p) for the vector y_t of the N log volatilities: each series on p lags of all series

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t] = 0, \quad \text{Cov}(\varepsilon_t) = \Sigma = (\sigma_{ij})$$

- ▶ c : N intercepts; A_l : $N \times N$ matrix, $(A_l)_{ij}$ = effect of $y_{j,t-l}$ on $y_{i,t}$; ε_t : shocks, serially uncorrelated but correlated across banks
- ▶ estimated by OLS equation by equation; $\hat{\Sigma}$: covariance of the residuals

- Lag order by BIC (B9): the smallest value of

$$\text{BIC}(p) = \ln \det \hat{\Sigma}_p + \frac{\ln T}{T} pN^2$$

- ▶ $\det \hat{\Sigma}_p$: generalised residual variance with p lags (fit); pN^2 slope coefficients; T : number of days (penalty)

- Moving-average form of a stable VAR (eigenvalues of the companion matrix inside the unit circle)

$$y_t = \mu + \sum_{h \geq 0} \Phi_h \varepsilon_{t-h}, \quad \Phi_0 = I, \quad \Phi_h = \sum_{l=1}^{\min(h,p)} A_l \Phi_{h-l}$$

- ▶ μ : mean of y_t ; Φ_h : effect after h days of a shock today; VAR(1): $\Phi_h = A^h$

Prerequisites for Today (13/23): Impulse Responses: Generalized and Cholesky

- Shocks are correlated, so “a shock to bank j alone” must be defined
 - ▶ e_j : the j -th unit vector; σ_{jj} : variance of $\varepsilon_{j,t}$
- **Generalized** response Pesaran & Shin (1998): shock j by one standard deviation, the other shocks at their conditional mean

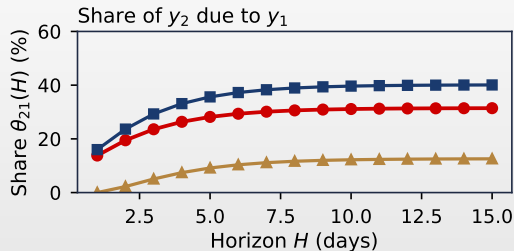
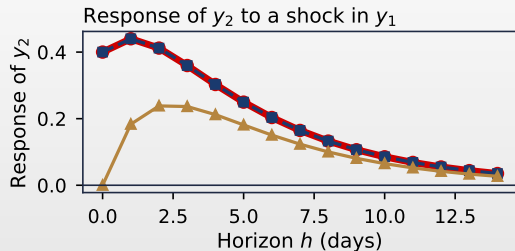
$$\psi_j^g(h) = \Phi_h \Sigma e_j / \sqrt{\sigma_{jj}}$$

- ▶ $\Sigma e_j / \sqrt{\sigma_{jj}} = \mathbb{E}[\varepsilon_t \mid \varepsilon_{j,t} = \sqrt{\sigma_{jj}}]$ for Normal shocks: the correlated banks move too; independent of the ordering
- **Cholesky** (orthogonalised) response: $\Sigma = PP^\top$, P lower triangular

$$\psi_j^c(h) = \Phi_h P e_j$$

- ▶ the first variable takes the whole common shock, the last none: the ordering decides
- ▶ for the variable ordered first the two responses coincide

Prerequisites for Today (14/23): Impulse Responses of a Simulated VAR(1)



● Generalized
 ■ Cholesky, variable 1 first
 ▲ Cholesky, variable 2 first

□ Two variables: $A = \begin{pmatrix} 0.7 & 0.1 \\ 0.2 & 0.6 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 1 \end{pmatrix}$

- ▶ left: with y_2 first, y_2 does not react on impact ($h = 0$); generalized and Cholesky with y_1 first coincide
- ▶ right: share of y_2 due to y_1 at $H = 10$: 40% or 12% by Cholesky ordering, 31% generalized

Prerequisites for Today (15/23): Forecast Error Variance Decomposition (FEVD)

- Forecast error of y_i at H steps and its variance

$$y_{t+H} - \mathbb{E}_t[y_{t+H}] = \sum_{h=0}^{H-1} \Phi_h \varepsilon_{t+H-h}, \quad \text{Var}_i(H) = \sum_{h=0}^{H-1} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i$$

- Generalized FEVD Pesaran & Shin (1998):** share of that variance due to shocks in j

$$\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i^\top \Phi_h \Sigma e_j)^2}{\text{Var}_i(H)}, \quad \tilde{\theta}_{ij}(H) = \frac{\theta_{ij}(H)}{\sum_{k=1}^N \theta_{ik}(H)}$$

- correlated shocks are counted for every bank they hit, so $\sum_j \theta_{ij} > 1$; the rows are normalised to sum to 1 [Diebold & Yilmaz \(2014\)](#)

- Cholesky FEVD:** the same with $P e_j$ in place of $\Sigma e_j / \sqrt{\sigma_{jj}}$

$$\theta_{ij}^c(H) = \sum_{h=0}^{H-1} (e_i^\top \Phi_h P e_j)^2 / \text{Var}_i(H)$$

- rows sum to 1 exactly, but the shares depend on the ordering (B5); with diagonal Σ both coincide

Prerequisites for Today (16/23): Connectedness Measures

- From the normalised table $\tilde{\theta}_{ij}$ (rows: receivers i , columns: sources j) Diebold & Yilmaz (2014)

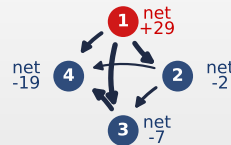
$$F_i = \sum_{j \neq i} \tilde{\theta}_{ij}, \quad T_j = \sum_{i \neq j} \tilde{\theta}_{ij}, \quad \text{Net}_i = T_i - F_i, \quad C = \frac{100}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$$

- F_i : received from others (row sum off the diagonal); T_j : transmitted to others (column sum)
- $\text{Net}_i > 0$: net transmitter; C : total connectedness, the share (%) of all forecast error variance that crosses banks
- Example: two banks, rows (0.7, 0.3) and (0.2, 0.8)
 - $F_1 = 0.3, F_2 = 0.2; T_1 = 0.2, T_2 = 0.3$
 - Net: bank 1 -0.1 (receiver), bank 2 $+0.1$ (transmitter); $C = \frac{100}{2}(0.3 + 0.2) = 25\%$

Prerequisites for Today (17/23): A Connectedness Table and Its Network

100 $\tilde{\theta}_{ij}(10)$ (%)				
Variance of bank i	1	2	3	4
1 -	67	17	11	6
2 -	27	54	12	8
3 -	22	17	55	7
4 -	13	10	16	60
To -	62	44	39	21
	1	2	3	4
	Shock in bank j			
	From			

Net pairwise spillovers



■ Net transmitter ($T_i - F_i > 0$)
 ■ Net receiver
 ➡ Arrow $j \rightarrow i: \tilde{\theta}_{ij} > \tilde{\theta}_{ji}$

- A simulated VAR(1) of four banks in which bank 1 drives the others with a lag; $H = 10$
 - ▶ Left: the table in %; last column F_i , last row T_j , corner $C = 41\%$
 - ▶ Right: bank 1 is the only net transmitter (+29); arrows show the net pairwise flows $\tilde{\theta}_{ij} - \tilde{\theta}_{ji}$

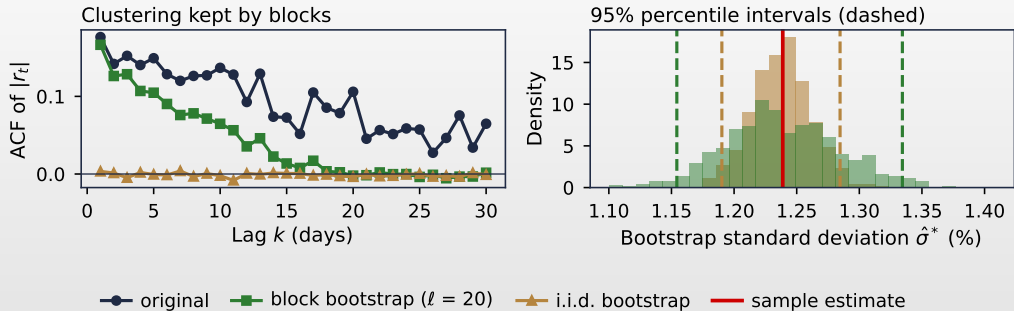
Prerequisites for Today (18/23): The Moving-Block Bootstrap

- Sampling uncertainty of an estimate $\hat{\theta}$ (e.g. ΔCoVaR , C) without a formula
 - ▶ $\hat{\theta}_b^*$: the estimate recomputed on bootstrap sample $b = 1, \dots, B$ ($B = 999$)
- **Moving blocks** [Künsch \(1989\)](#): glue $\lceil n/\ell \rceil$ blocks $(X_s, \dots, X_{s+\ell-1})$ with random starts s
 - ▶ $\ell = 20$ days keeps the volatility clustering inside each block; the i.i.d. bootstrap ($\ell = 1$) destroys it
 - ▶ whole days (all banks together) are resampled, so cross-bank dependence is kept
 - ▶ VAR (B9): blocks of residuals $\hat{\varepsilon}_t$, then $y_t^* = \hat{c} + \sum_I \hat{A}_I y_{t-I}^* + \varepsilon_t^*$
- What we report

$$\text{CI}_{95\%} = [\hat{\theta}_{(2.5\%)}^*, \hat{\theta}_{(97.5\%)}^*], \quad \text{se} = \text{sd}(\hat{\theta}_1^*, \dots, \hat{\theta}_B^*), \quad z = \hat{\theta}/\text{se}$$

 - ▶ percentile interval; bootstrap standard error; for $H_0: \theta = 0$, $p = 2(1 - \Phi(|z|))$, reject at 5% if $|z| > 1.96$

Prerequisites for Today (19/23): Blocks against i.i.d. Resampling



□ A simulated GARCH series ($T = 2000$ days) with volatility clustering

- ▶ Left: block resamples keep the autocorrelation of $|r_t|$ up to about $\ell = 20$ lags; i.i.d. resamples have none
- ▶ Right: the block interval for the standard deviation is about twice as wide ($[1.15, 1.33]$ against $[1.19, 1.29]$): the i.i.d. bootstrap overstates the precision

Prerequisites for Today (20/23): Many Tests at Once: Holm, Wald and Spearman

- **Step-down correction Holm (1979):** m tests, p -values sorted $p_{(1)} \leq \dots \leq p_{(m)}$

$$\text{reject } H_{(k)} \text{ while } p_{(k)} \leq \frac{0.05}{m - k + 1}, \quad k = 1, 2, \dots; \text{ stop at the first failure}$$

- ▶ family-wise error rate (FWER) $\leq 5\%$: probability of at least one false rejection, valid under any dependence

- **Wald test** that m contributions are equal (B1, B2)

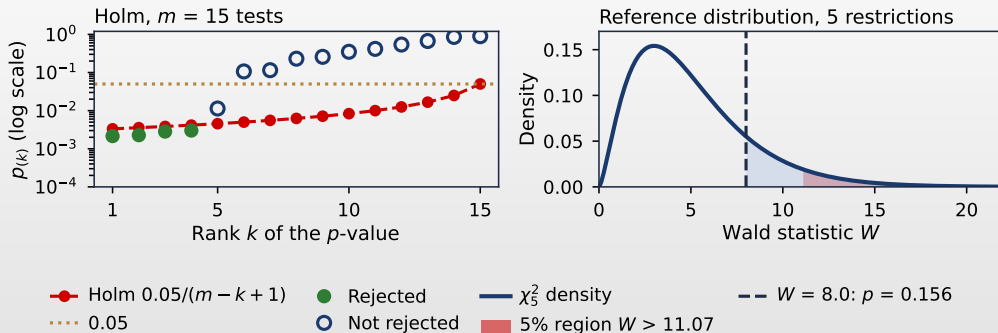
$$W = \hat{d}^\top \hat{V}^{-1} \hat{d} \sim \chi_{m-1}^2 \text{ under } H_0$$

- ▶ $\hat{d}$: the $m - 1$ differences $\widehat{\Delta \text{CoVaR}}_i - \widehat{\Delta \text{CoVaR}}_m$; $\hat{V}$: their covariance across common bootstrap draws
- ▶ uses the correlation of the estimates, which separate intervals ignore; reject if W exceeds the 95% quantile of χ_{m-1}^2

- **Spearman** rank correlation of two rankings of n banks (B2)

- ▶ $r_S = 1 - 6 \sum_i d_i^2 / (n(n^2 - 1))$, d_i : difference of the two ranks of bank i ; 1: same order, -1: reversed

Prerequisites for Today (21/23): Holm Thresholds and the χ^2 Reference Distribution



- Left: 15 simulated p -values (4 from false nulls); 5 are below 0.05, Holm keeps 4
 - the threshold rises from $0.05/15 = 0.0033$ to 0.05; testing stops at the first point above its threshold
- Right: χ^2_5 , the Wald reference for six banks; 5% critical value 11.07
 - p -value = area to the right of the observed W : $W = 8.0$ gives $p = 0.156$, no rejection

Prerequisites for Today (22/23): Extension Packs: CoVaR Variants and Tests

- CoVaR conditional on losses **at least** VaR [Girardi & Ergün \(2013\)](#) (A2, B10)

$$P(X^{sys} \leq -\text{CoVaR}_\alpha^\leq \mid X_i \leq q_\alpha(X_i)) = \alpha$$

- ▶ conditions on the whole tail of the bank, not on one point; larger than CoVaR when $\rho > 0$

- Kupiec test of a hit rate [Banulescu-Radu, Hurlin, Leymarie & Scaillet \(2021\)](#): x hits in n days, nominal rate π_0 , $\hat{\pi} = x/n$

$$LR = -2[x \ln \pi_0 + (n - x) \ln(1 - \pi_0) - x \ln \hat{\pi} - (n - x) \ln(1 - \hat{\pi})] \sim \chi_1^2$$

- ▶ reject at 5% if $LR > 3.84$; for a joint CoVaR event $\pi_0 = \alpha^2$; power: probability of rejecting a wrong rate

- Newey–West standard errors [Newey & West \(1987\)](#) (B6): OLS with autocorrelated errors

$$\hat{\Omega} = \hat{\Gamma}_0 + \sum_{l=1}^L \left(1 - \frac{l}{L+1}\right) (\hat{\Gamma}_l + \hat{\Gamma}_l^\top), \quad \hat{\Gamma}_l = \frac{1}{T} \sum_t \hat{u}_t \hat{u}_{t-l} x_t x_{t-l}^\top$$

- ▶ $\hat{u}_t$: residual, x_t : regressors, L : lags; $\widehat{\text{Var}}(\hat{\beta}) = (X^\top X)^{-1} T \hat{\Omega} (X^\top X)^{-1}$; needed for overlapping rolling windows

Prerequisites for Today (23/23): Extension Packs: GARCH–DCC and Stress Tests

- GJR–GARCH(1,1) (B3, B4): bad news raise the variance more

$$\sigma_t^2 = \omega + (a + \gamma 1\{r_{t-1} < 0\})r_{t-1}^2 + b\sigma_{t-1}^2$$

- ▶ $\omega > 0$; a : reaction to news; $\gamma > 0$: extra reaction to losses; b : memory; persistence $a + \gamma/2 + b < 1$

- DCC(1,1): a time-varying correlation of the standardised residuals ε_t

- ▶ $Q_t = (1 - a - b)\bar{Q} + a\varepsilon_{t-1}\varepsilon_{t-1}^\top + bQ_{t-1}$, $\rho_t = Q_{12,t}/\sqrt{Q_{11,t}Q_{22,t}}$; $\bar{Q}$: sample covariance of ε_t

- Reverse stress test (A7, A8): the most likely factor move f with loss ℓ^* , $f \sim N(0, \Sigma)$

- ▶ maximise the density, i.e. minimise the squared Mahalanobis distance $d^2 = f^\top \Sigma^{-1} f$, subject to $-b^\top f = \ell^*$ (b : exposures)
- ▶ d : distance in standard deviations that accounts for correlation; Student- t factors (A8): same direction, heavier tail probability

Notation Used in Every Exercise

▣ Returns

- ▶ X_i : daily log return of bank i (%); X^{sys} : equal-weighted return of the **other** banks of its region; X_m : market index
- ▶ α : tail probability; $q_\alpha(X)$: α -quantile; $\text{VaR}_\alpha = -q_\alpha$; VaR, CoVaR, MES are positive losses

▣ Contribution and exposure

- ▶ $q_\alpha(X^{sys} | X_i) = a_\alpha + b_\alpha X_i$; $\Delta\text{CoVaR}_\alpha = \hat{b}_\alpha(\hat{q}_{50\%}(X_i) - \hat{q}_\alpha(X_i))$ [Adrian & Brunnermeier \(2016\)](#)
- ▶ $\text{MES}_{i,\alpha} = -\mathbb{E}[X_i | X_m \leq q_\alpha(X_m)]$ [Acharya et al. \(2017\)](#); $\text{SRISK} = kD - (1 - k)(1 - \text{LRMES})W$,
 $L^* = 1 + (1 - k)(1 - \text{LRMES})/k$ [Brownlees & Engle \(2017\)](#)

▣ Connectedness [Diebold & Yilmaz \(2014\)](#)

- ▶ $\tilde{\theta}_{ij}$: share of the H -day forecast error variance of i due to j ; F_i , T_j , net $T_i - F_i$, total C as in the primer

▣ Uncertainty

- ▶ Moving-block bootstrap, blocks of 20 days, 999 draws [Künsch \(1989\)](#)
- ▶ Holm correction [Holm \(1979\)](#): the k -th smallest of m p -values is compared with $0.05/(m - k + 1)$, stopping at the first non-rejection

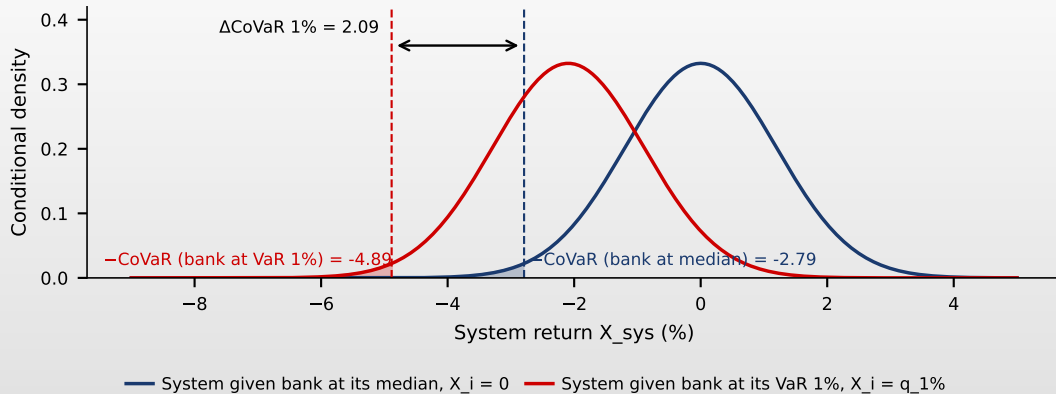
A1 [Solved]: Deriving CoVaR under the Normal Distribution — Task

- ▣ **Question:** How much does a bank in distress add to the tail loss of the system when returns are Normal?
- ▣ **Data and assumptions**
 - ▶ (X_i, X^{sys}) : daily returns (%) of bank i and of the system, bivariate Normal with zero means, standard deviations σ_i, σ_{sys} and correlation $\rho > 0$
 - ▶ Distress of the bank: $X_i = q_\alpha(X_i) = \sigma_i z_\alpha$, with $z_\alpha = \Phi^{-1}(\alpha) < 0$; median state: $X_i = 0$; CoVaR is minus the α -quantile of X^{sys} , a positive loss [Adrian & Brunnermeier \(2016\)](#)
 - ▶ Numbers for (c): $\sigma_i = 2\%$, $\sigma_{sys} = 1.5\%$, $\rho = 0.6$, $\alpha = 1\%$
- ▣ **Tasks**
 - (a) Project X^{sys} on X_i and derive the conditional distribution of X^{sys} given $X_i = x$ (mean and variance).
 - (b) Derive CoVaR_α (given $X_i = \sigma_i z_\alpha$), the median-state CoVaR (given $X_i = 0$) and $\Delta\text{CoVaR}_\alpha$ as their difference; show that $\Delta\text{CoVaR}_\alpha$ does not depend on σ_i .
 - (c) Compute VaR 1% of the bank, CoVaR 1%, the median-state CoVaR and ΔCoVaR 1% for the numbers above.
- ▣ **Report**
 - ▶ The conditional mean and standard deviation, the three closed forms of (b) and the four numbers of (c), with every intermediate product
- ▣ **Interpretation**
 - ▶ A bank twice as volatile, with the same ρ , has the same ΔCoVaR : what does this say about ranking banks by their own VaR?
- ▣ **Notebook:** section A1

A1 [Solved]: Solution (1/3)

- (a) Projection: $X^{sys} = \beta X_i + U$ with $\beta = \text{Cov}(X^{sys}, X_i) / \sigma_i^2 = \rho \sigma_{sys} / \sigma_i$ and U Normal, independent of X_i .
- Residual variance: $\text{Var}(U) = \sigma_{sys}^2 - \beta^2 \sigma_i^2 = \sigma_{sys}^2 (1 - \rho^2)$.
- Result: $X^{sys} \mid X_i = x \sim N\left(\rho \frac{\sigma_{sys}}{\sigma_i} x, \sigma_{sys}^2 (1 - \rho^2)\right)$.
- (b) The α -quantile given x is $\rho \frac{\sigma_{sys}}{\sigma_i} x + \sigma_{sys} \sqrt{1 - \rho^2} z_\alpha$.
- At $x = \sigma_i z_\alpha$: $\text{CoVaR}_\alpha = -\sigma_{sys} z_\alpha (\rho + \sqrt{1 - \rho^2})$; at $x = 0$: $\text{CoVaR}_\alpha^{50\%} = -\sigma_{sys} \sqrt{1 - \rho^2} z_\alpha$.
- Difference: $\Delta \text{CoVaR}_\alpha = -\rho \sigma_{sys} z_\alpha$; σ_i cancels because the distress state x grows with σ_i while the slope falls with $1/\sigma_i$.

A1 [Solved]: Solution (2/3)



A1 [Solved]: Solution (3/3)

- Blue: system given the bank at its median; red: given the bank at its VaR 1%; shaded: the 1% tails; the gap between the dashed lines is $\Delta\text{CoVaR 1\%}$
- (c) $z_{1\%} = -2.326$; VaR 1% of the bank:
 $-\sigma_i z_{1\%} = 2 \cdot 2.326 = 4.65\%$.
- Slope $\rho\sigma_{\text{sys}}/\sigma_i = 0.6 \cdot 1.5/2 = 0.45$; conditional standard deviation $1.5\sqrt{1 - 0.36} = 1.20\%$.
- CoVaR 1%:
 $0.45 \cdot 4.65 + 1.20 \cdot 2.326 = 2.094 + 2.792 = 4.885\%$.
- Median state: $1.20 \cdot 2.326 = 2.792\%$; so
 $\Delta\text{CoVaR}_{1\%} = 0.6 \cdot 1.5 \cdot 2.326 = 2.094\%$, the first term above.
- Interpretation: when the bank moves from its median to its VaR, the system loses about 2.09 extra points; doubling σ_i doubles the VaR but leaves ΔCoVaR unchanged.
- Own risk and contribution to systemic risk are different quantities: a ranking by VaR is not a systemic ranking.

A3 [Solved]: MES as an Euler Contribution — Task (1/2)

- **Question:** Why do the MES of the assets add up exactly to the ES of the portfolio?
- **Data and assumptions**
 - ▶ Portfolio $X_p = \sum_i w_i X_i$ of daily returns with a continuous distribution; $q_\alpha(w)$ is its α -quantile
 - ▶ $ES_\alpha(X_p) = -\frac{1}{\alpha} \mathbb{E}[X_p \mathbf{1}\{X_p \leq q_\alpha\}]$ and $MES_{i,\alpha} = -\mathbb{E}[X_i \mid X_p \leq q_\alpha]$ [Acharya et al. \(2017\)](#)
 - ▶ Numbers for (c): jointly Normal, zero means, $w = (0.5, 0.5)$, $\sigma = (2\%, 3\%)$, $\rho = 0.5$, $\alpha = 5\%$
- **Tasks**
 - (a) Differentiate ES_α with respect to w_i , including the effect of w_i on the quantile, to show $\partial ES_\alpha / \partial w_i = MES_{i,\alpha}$; then use homogeneity to show $ES_\alpha = \sum_i w_i MES_{i,\alpha}$.
 - (b) For jointly Normal returns with zero means, show that $MES_{i,\alpha} = \beta_{i,p} ES_\alpha(X_p)$ with $\beta_{i,p} = \text{Cov}(X_i, X_p) / \sigma_p^2$ and $ES_\alpha(X_p) = \sigma_p \varphi(z_\alpha) / \alpha$.
 - (c) Compute σ_p , $ES_{5\%}$, both betas and both MES for the numbers above, and check the Euler sum.

A3 [Solved]: MES as an Euler Contribution — Task (2/2)

□ Report

- ▶ σ_p , $\varphi(z_{5\%})/0.05$, $ES_{5\%}$, β , MES and the check $\sum_i w_i MES_i = ES_{5\%}$

□ Interpretation

- ▶ Asset 2 has half the total portfolio weight: what share of the ES does it carry?
- ▶ Why is MES, not volatility, the right key for allocating tail risk?

□ Notebook: section A3

A3 [Solved]: Solution (1/2)

- (a) Write $\alpha \text{ES}_\alpha = -\mathbb{E}[X_p \mathbf{1}\{X_p \leq q(w)\}]$; differentiating under the expectation gives the direct term $-\mathbb{E}[X_i \mathbf{1}\{X_p \leq q\}]$.
- The boundary term is $-q \frac{\partial}{\partial w_i} P(X_p \leq q(w))$: the mass that crosses the boundary, where $X_p = q$.
- $P(X_p \leq q(w)) = \alpha$ for every w , so this derivative is zero and $\partial \text{ES}_\alpha / \partial w_i = -\frac{1}{\alpha} \mathbb{E}[X_i \mathbf{1}\{X_p \leq q\}] = \text{MES}_{i,\alpha}$.
- Homogeneity: $\text{ES}_\alpha(tX_p) = t \text{ES}_\alpha(X_p)$, so Euler's theorem gives $\text{ES}_\alpha = \sum_i w_i \partial \text{ES}_\alpha / \partial w_i = \sum_i w_i \text{MES}_{i,\alpha}$ [Gouriéroux, Laurent & Scaillet \(2000\)](#).
- (b) Normal regression: $\mathbb{E}[X_i | X_p] = \beta_{i,p} X_p$, so $\text{MES}_{i,\alpha} = -\beta_{i,p} \mathbb{E}[X_p | X_p \leq q] = \beta_{i,p} \text{ES}_\alpha(X_p)$.
- For $X_p \sim N(0, \sigma_p^2)$, $\text{ES}_\alpha = \sigma_p \varphi(z_\alpha) / \alpha$; and $\sum_i w_i \beta_{i,p} = w^\top \Sigma w / \sigma_p^2 = 1$ confirms the Euler sum.

A3 [Solved]: Solution (2/2)

- (c) Covariance $\rho\sigma_1\sigma_2 = 0.5 \cdot 2 \cdot 3 = 3$; $\sigma_p^2 = 0.25 \cdot 4 + 2 \cdot 0.25 \cdot 3 + 0.25 \cdot 9 = 4.75$, so $\sigma_p = 2.179\%$.
- $\varphi(z_{5\%})/0.05 = 2.063$, so $ES_{5\%} = 2.179 \cdot 2.063 = 4.50\%$.
- $\text{Cov}(X_i, X_p) = (\Sigma w)_i = (3.5, 6.0)$; $\beta = (3.5, 6.0)/4.75 = (0.737, 1.263)$.
- $MES = \beta \cdot 4.50 = (3.31, 5.68)\%$; check: $0.5(3.31 + 5.68) = 4.50\% = ES_{5\%}$.
- Interpretation: with half the weight, asset 2 carries 63% of the ES, because it moves more with the portfolio in its tail.
- MES splits the tail loss additively and uses the co-movement with the portfolio, which volatility alone ignores.

A5 [Solved]: Deriving the Generalized FEVD — Task (1/2)

- **Question:** How is the forecast error variance of one bank split between shocks to all banks?
- **Data and assumptions**
 - ▶ Stable VAR(1) $y_t = Ay_{t-1} + \varepsilon_t$ (eigenvalues of A inside the unit circle), ε_t serially uncorrelated, zero-mean and jointly Normal with positive definite $\text{Cov}(\varepsilon_t) = \Sigma = (\sigma_{ij})$; moving-average matrices $\Phi_h = A^h$; e_j is the j -th unit vector
 - ▶ Generalized impulse [Pesaran & Shin \(1998\)](#): a one-standard-deviation shock $\delta_j = \sqrt{\sigma_{jj}}$ to variable j , the other shocks at their conditional expectation
 - ▶ Numbers for (b): $A = \begin{pmatrix} 0.5 & 0.2 \\ 0.1 & 0.4 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$
- **Tasks**
 - (a) Show that $\mathbb{E}[\varepsilon_t \mid \varepsilon_{jt} = \delta_j] = \Sigma e_j / \sqrt{\sigma_{jj}}$ and derive the share $\theta_{ij}(H)$ of the H -step forecast error variance of i due to shocks in j .
 - (b) Compute θ , its row sums, the row-normalised table $\tilde{\theta}$ and total connectedness C at $H = 1$ and $H = 2$.
 - (c) Compute the Cholesky decomposition at $H = 1$ with variable 1 first, and show that with Σ diagonal the generalized and Cholesky decompositions coincide.

A5 [Solved]: Deriving the Generalized FEVD — Task (2/2)

□ Report

- ▶ θ and $\tilde{\theta}$ with row sums and C at $H = 1, 2$, with $A\Sigma$ and $\text{diag}(A\Sigma A^\top)$; the Cholesky table

□ Interpretation

- ▶ Why do the rows of the generalized table not sum to one?
- ▶ What does the Cholesky order decide?

□ Notebook: section A5

A5 [Solved]: Solution (1/3)

- (a) For Normal shocks $\mathbb{E}[\varepsilon_t | \varepsilon_{jt}] = \Sigma e_j \varepsilon_{jt} / \sigma_{jj}$; at $\varepsilon_{jt} = \sqrt{\sigma_{jj}}$ this is $\Sigma e_j / \sqrt{\sigma_{jj}}$.
- Response of variable i after h steps: $e_i^\top \Phi_h \Sigma e_j / \sqrt{\sigma_{jj}}$; the H -step forecast error variance of i is $\sum_{h < H} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i$.
- Share: $\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h < H} (e_i^\top \Phi_h \Sigma e_j)^2}{\sum_{h < H} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i}$.
- Normalisation $\tilde{\theta}_{ij} = \theta_{ij} / \sum_j \theta_{ij}$ and total connectedness $C = \frac{100}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$ Diebold & Yilmaz (2014).

A5 [Solved]: Solution (2/3)

- (b) $H = 1$ ($\Phi_0 = I$): $\theta_{ij} = \sigma_{ij}^2 / (\sigma_{ii}\sigma_{jj})$, so $\theta = \begin{pmatrix} 1 & 0.25 \\ 0.25 & 1 \end{pmatrix}$, row sums 1.25.
- Normalised: $\tilde{\theta} = \begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{pmatrix}$, so $C = 100(0.2 + 0.2)/2 = 20.0\%$.
- $H = 2$ adds $\Phi_1 = A$: $A\Sigma = \begin{pmatrix} 0.60 & 0.45 \\ 0.30 & 0.45 \end{pmatrix}$, $\text{diag}(A\Sigma A^\top) = (0.39, 0.21)$; denominators 1.39 and 1.21.
- Numerators $\sigma_{ij}^2 + (A\Sigma)_{ij}^2$: row 1 (1.3600, 0.4525), row 2 (0.3400, 1.2025); so $\theta = \begin{pmatrix} 0.978 & 0.326 \\ 0.281 & 0.994 \end{pmatrix}$, row sums 1.304 and 1.275.
- Normalised off-diagonal shares 0.250 and 0.220, so $C = 100(0.250 + 0.220)/2 = 23.5\%$.

A5 [Solved]: Solution (3/3)

- (c) Cholesky with variable 1 first: $P = \begin{pmatrix} 1 & 0 \\ 0.5 & \sqrt{0.75} \end{pmatrix}$, $PP^\top = \Sigma$; at $H = 1$ the shares are the squared entries of the rows of P : $(1, 0)$ and $(0.25, 0.75)$.
- The whole common shock is given to variable 1, the first in the order.
- Diagonal Σ : $\Sigma e_j / \sqrt{\sigma_{jj}} = \sqrt{\sigma_{jj}} e_j = P e_j$ with $P = \Sigma^{1/2}$ the diagonal Cholesky factor: the impulses coincide and the rows already sum to one.
- Interpretation: rows of θ exceed one because a correlated shock is counted for both variables; normalisation fixes the sum without imposing an order.
- With Cholesky the order, not the data, decides who receives the common shock; the lagged links of A raise C from 20.0% to 23.5% between $H = 1$ and $H = 2$.

B1 [Solved]: ΔCoVaR 5% of US Banks, with Tests — Task (1/2)

- **Question:** Do the six US banks differ in their contribution to systemic risk?
- **Data and assumptions**
 - ▶ Adjusted closes of six US banks (course dataset): JPMorgan, Bank of America, Citigroup, Goldman Sachs, Morgan Stanley, Wells Fargo
 - ▶ Sample: 5 January 2010 – 18 September 2026, 3 960 trading days common to all 13 banks and the four indices (prices aligned on common days first, then daily log returns in %)
 - ▶ System of bank i : the equal-weighted portfolio of the other five banks [Adrian & Brunnermeier \(2016\)](#); level $\alpha = 5\%$, about 198 tail days
- **Tasks**
 - (a) For each bank, estimate the quantile regression $q_{5\%}(X_t^{\text{sys}} \mid X_{i,t}) = a_{5\%} + b_{5\%}X_{i,t}$ and compute VaR 5%, CoVaR 5% and $\Delta\text{CoVaR}_{5\%} = \hat{b}_{5\%}(\hat{q}_{50\%} - \hat{q}_{5\%})$.
 - (b) Build 95% percentile intervals for ΔCoVaR with a moving-block bootstrap (blocks of 20 days) [Künsch \(1989\)](#) and with an i.i.d. bootstrap, 999 resamples each.
 - (c) Test $H_0 : b_{5\%} = b_{50\%}$ for each bank with $z = (\hat{b}_{5\%} - \hat{b}_{50\%})/\text{se}$ (block-bootstrap se) at the 5% level, with and without the Holm correction [Holm \(1979\)](#).
 - (d) Test equal ΔCoVaR for the 15 pairs on joint draws (the same resampled days for all banks) with Holm, and for all six banks jointly with a Wald test (χ^2_5).

B1 [Solved]: ΔCoVaR 5% of US Banks, with Tests — Task (2/2)

□ Report

- ▶ A table with $\hat{b}_{5\%}$, ΔCoVaR , both intervals, $\hat{b}_{5\%} - \hat{b}_{50\%}$ and its p -value; a chart of the two intervals per bank
- ▶ The number of rejections in (c) and (d), raw and after Holm; the Wald statistic, its p -value and the decision

□ Interpretation

- ▶ Why is the block interval wider than the i.i.d. one?
- ▶ Can overlapping marginal intervals hide a significant difference?
- ▶ Why is bank i left out of its own system?

□ Notebook: section B1

B1 [Solved]: Solution (1/4)

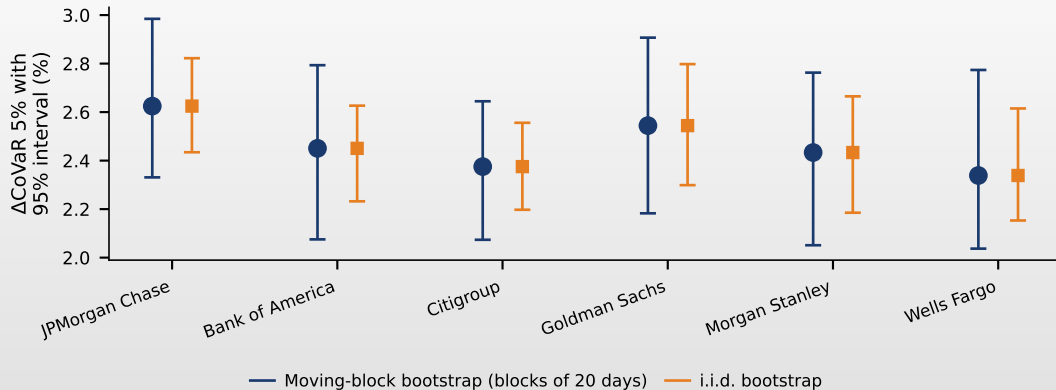
- Step 1: for each bank the system return is $X_t^{\text{sys}} = \frac{1}{5} \sum_{j \neq i} X_{j,t}$; estimate the quantile regressions at 5% and at 50%.
- Step 2: plug in the empirical quantiles: $\text{CoVaR}_{5\%} = -(\hat{\alpha} + \hat{b}\hat{q}_{5\%})$ and $\Delta\text{CoVaR}_{5\%} = \hat{b}(\hat{q}_{50\%} - \hat{q}_{5\%})$.
- Example, JPMorgan: $\text{VaR } 5\% = 2.66\%$, $\hat{q}_{50\%} - \hat{q}_{5\%} = 2.74$, $\hat{b}_{5\%} = 0.96$, so $\Delta\text{CoVaR}_{5\%} = 0.96 \cdot 2.74 = 2.62\%$; $\text{CoVaR } 5\% = 3.81\%$.
- Step 3: draw 999 samples of 20-day blocks (block bootstrap) and 999 samples of single days (i.i.d.), re-estimate on each and take the 2.5% and 97.5% percentiles.
- Step 4: in each block draw compute all six $\hat{b}_{5\%} - \hat{b}_{50\%}$ and all six ΔCoVaR on the same days; the standard deviations of the draws are the standard errors, their covariance feeds the Wald test.

B1 [Solved]: Solution (2/4)

	$\hat{b}_{5\%}$	ΔCoVaR	block 95% interval	i.i.d. 95% interval	$\hat{b}_{5\%} - \hat{b}_{50\%}$	p
JPMorgan	0.96	2.62	[2.33, 2.98]	[2.43, 2.82]	-0.002	0.878
Bank of America	0.76	2.45	[2.08, 2.79]	[2.23, 2.63]	-0.017	0.425
Citigroup	0.72	2.38	[2.07, 2.64]	[2.20, 2.56]	-0.040	0.019
Goldman Sachs	0.89	2.54	[2.18, 2.91]	[2.30, 2.80]	0.044	0.048
Morgan Stanley	0.72	2.43	[2.05, 2.76]	[2.19, 2.66]	-0.009	0.746
Wells Fargo	0.81	2.34	[2.04, 2.77]	[2.15, 2.61]	0.032	0.221

- (a) All six contributions lie between 2.34% and 2.62% of extra system loss; slopes lie between 0.72 and 0.96.
- (b) Block intervals are wider (JPMorgan: width 0.65 against 0.39 points): tail days cluster in crises, and the i.i.d. bootstrap breaks the clusters apart.

B1 [Solved]: Solution (3/4)



B1 [Solved]: Solution (4/4)

- (c) Citigroup: $z = -0.040/0.017 = -2.34$, $p = 0.019$, rejected at 5%; JPMorgan: $z = -0.15$, $p = 0.878$, not rejected; overall, the null is rejected for 2 of 6 banks, and for 0 after Holm.
- (d) 2 of 15 pairs differ at 5%, 0 after Holm; Wald $\chi^2_5 = 14.2$, $p = 0.014 < 0.05$: we reject that the six contributions are all equal.
- Interpretation: the joint test sees differences that overlapping marginal intervals hide, because it uses the positive correlation of the estimates across banks.
- Bank i is left out of its system: otherwise X_i would appear on both sides of the regression and add a mechanical self-contribution to $\hat{b}$ (with w the weight of the bank in the system and b the slope on the rest, the slope becomes $w + (1 - w)b$).

B5 [Solved]: How Robust Is the Connectedness Index? — Task

- **Question:** How much do modelling choices move the connectedness index?
- **Data and assumptions**
 - ▶ Daily high and low prices of the 13 banks (course dataset; six US, five European, Banca Transilvania and BRD), 2010–2026, common days
 - ▶ Parkinson volatility $\hat{\sigma}_t = \sqrt{252 (\ln H_t - \ln L_t)^2 / (4 \ln 2)}$, used in logs; order of the variables: US banks first, Romanian banks last
 - ▶ Generalized connectedness [Diebold & Yilmaz \(2014\)](#) with the decomposition of [Pesaran & Shin \(1998\)](#) (A5)
- **Tasks**
 - (a) Estimate $\text{VAR}(p)$ by OLS for $p = 1, \dots, 4$ and compute total connectedness C at $H = 5, 10, 20$ (12 values).
 - (b) With $p = 1$, $H = 10$, compute C and the summed net position of the six US banks under the generalized decomposition and under Cholesky with US banks first and with Romanian banks first.
- **Report**
 - ▶ The 4×3 table and chart of (a); the three totals and three net positions of (b)
- **Interpretation**
 - ▶ Which choice moves the index most?
 - ▶ Why does the Cholesky order change the net position of US banks?
- **Notebook:** section B5

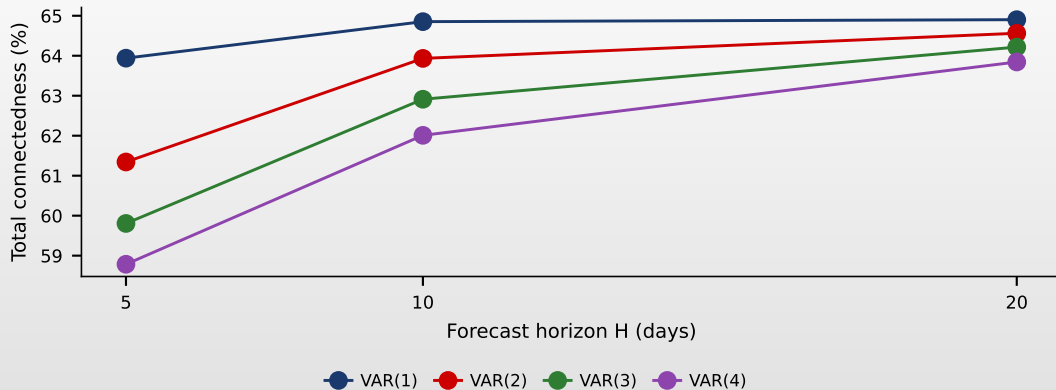
B5 [Solved]: Solution (1/3)

- Steps: (1) log Parkinson volatilities; (2) $\text{VAR}(p)$ with a constant by OLS and the residual covariance $\hat{\Sigma}$; (3) Φ_h for $h < H$; (4) generalized shares, rows normalised; (5) $C = \frac{100}{13} \sum_{i \neq j} \tilde{\theta}_{ij}$.

C (%)	$H = 5$	$H = 10$	$H = 20$
VAR(1)	63.9	64.9	64.9
VAR(2)	61.3	63.9	64.6
VAR(3)	59.8	62.9	64.2
VAR(4)	58.8	62.0	63.8

- (a) The index moves between 58.8% ($p = 4, H = 5$) and 64.9% ($p = 1, H = 20$): a range of 6.1 points over all 12 choices.

B5 [Solved]: Solution (2/3)



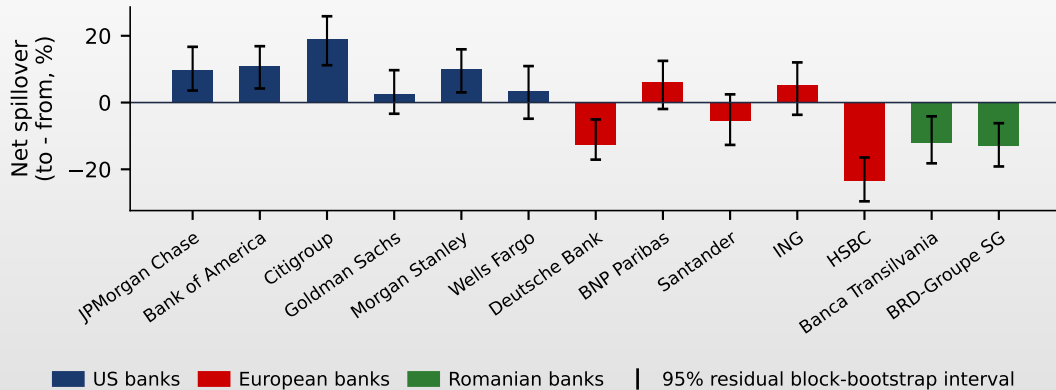
B5 [Solved]: Solution (3/3)

- (b) Totals: Cholesky 44.1% (US first) and 43.5% (Romanian first) against generalized 64.9%.
- Net position of the US banks: +65.9 (US first), -96.0 (Romanian first), +53.2 (generalized).
- Interpretation: the decomposition moves the index most; Cholesky gives the common shock to the variables ordered first, so the order, not the data, decides who transmits: use the generalized decomposition.

B9 [Solved]: Which Net Spillovers Are Identified? — Task

- **Question:** Which transmitter and receiver labels of the network survive estimation error?
- **Data and assumptions**
 - ▶ Log Parkinson volatilities of the 13 banks (B5), 2010–2026; VAR order $p = 2$ chosen by BIC (up to 5 lags), $H = 10$, generalized decomposition [Diebold & Yilmaz \(2014\)](#)
 - ▶ Window of the 2020 peak of the rolling index: the 250 days ending 2020-08-27, VAR(2) as in the rolling index
- **Tasks**
 - (a) Build 999 bootstrap samples: resample the VAR residuals in blocks of 20 days, rebuild the series with the estimated VAR, re-estimate the VAR and recompute the connectedness table.
 - (b) Report 95% percentile intervals for C and for the 13 net positions; call a bank a transmitter (receiver) only if its interval lies above (below) zero.
 - (c) Repeat (a)–(b) for C in the peak window and compare the bootstrap mean with the estimate.
- **Report**
 - ▶ C with its interval; a chart of the 13 net positions with intervals; the identified banks; the peak-window estimate, interval and bootstrap mean
- **Interpretation**
 - ▶ Which “transmitter” and “receiver” labels survive?
 - ▶ Why is the bootstrap mean of C in the short window different from the estimate?
- **Notebook:** section B9

B9 [Solved]: Solution (1/3)



B9 [Solved]: Solution (2/3)

- Steps: (1) VAR(2) by OLS, residuals $\hat{\varepsilon}_t$; (2) resample 20-day blocks of $\hat{\varepsilon}_t$; (3) rebuild $y_t^* = \hat{c} + \sum_{l=1}^2 \hat{A}_l y_{t-l}^* + \varepsilon_t^*$ ($\hat{c}$, $\hat{A}_l$: estimated VAR constants and coefficients, ε_t^* : resampled residuals) from the first 2 observed values; (4) re-estimate, recompute the table; (5) percentiles over 999 draws.
- Net positions with 95% intervals; a label is identified only if its interval excludes zero.

B9 [Solved]: Solution (3/3)

- (a)–(b) Full sample: $C = 63.9\%$, interval $[59.9, 65.8]$.
- 8 of 13 net positions exclude zero: JPMorgan, Bank of America, Citigroup, Morgan Stanley, Deutsche Bank, HSBC, Banca Transilvania, BRD-Groupe SG; the other five labels are not identified.
- (c) 2020 peak (250 days to 2020-08-27): $C = 79.3\%$, interval $[65.9, 83.0]$, bootstrap mean 75.8% , i.e. -3.5 points from the estimate.
- Interpretation: the gap between the bootstrap mean and the estimate is a bootstrap estimate of finite-sample bias: in short windows the index is biased as well as noisy; sampling noise in the OLS coefficients is one plausible source [Demirer et al. \(2018\)](#).
- Report the interval next to every net number, and draw only identified labels on a network chart.

 MFM_ch18_spillover_uncertainty

C2 [Proposed]: Critique an AI Answer — the Answer

- **Student prompt:** “Bank i and the system are jointly Normal with zero means: $\sigma_i = 2.5\%$, $\sigma_{sys} = 1.2\%$, $\rho = 0.5$. Compute CoVaR 1%, ΔCoVaR 1%, MES 5% (the system is the market) and SRISK with $W = 100$, $D = 1100$, $k = 8\%$, LRMES = 40%.”
 - ▶ W : market value of equity; D : book value of debt; k : prudential capital ratio
- **Answer of the AI assistant**
 - ▶ $\text{VaR}_i = 2.5 \cdot 2.326 = 5.82\%$; $X^{sys} \mid X_i = x \sim N(0.24x, 1.04^2)$
 - ▶ CoVaR conditions on the bank at its VaR, $X_i = \text{VaR}_i = 5.82$: the 1% quantile is $0.24 \cdot 5.82 - 1.04 \cdot 2.326 = -1.02$, so CoVaR = 1.02%
 - ▶ Median state: $1.04 \cdot 2.326 = 2.42\%$; $\Delta\text{CoVaR} = 1.02 - 2.42 = -1.40\%$: the bank **reduces** systemic risk
 - ▶ MES is the expected loss of the bank in its own 5% tail: $\text{MES} = 2.5 \cdot \varphi(1.645)/0.05 = 5.16\%$
 - ▶ $\text{SRISK} = kD - (1 - \text{LRMES})W = 88 - 60 = 28$

C2 [Proposed]: Critique an AI Answer — Task

- The answer on the previous slide contains **three** errors common in calculations of systemic-risk measures; the rest is correct
 - ▶ Model: A1, A3 and the SRISK formula of the primer; notebook section C2; conventions of this seminar: α is a tail probability, VaR, CoVaR and MES are positive losses
- Your tasks
 - ▶ Find each error: which step, and why it is wrong
 - ▶ Say how you would detect it without knowing the answer: a sign check, a limiting case, a closed form
 - ▶ Give the corrected numbers
 - ▶ Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

B2 [Proposed]: ΔCoVaR of European Banks and the VaR Ranking — Task (1/2)

□ Questions

- ▶ Can we name one European bank “most systemic”?
- ▶ Does its VaR tell us which?

□ Data and assumptions

- ▶ Adjusted closes of five European banks (course dataset): Deutsche Bank, BNP Paribas, Santander, ING, HSBC; same sample and returns as B1
- ▶ System of each bank: the equal-weighted portfolio of the other four; levels $\alpha = 1\%$ (about 40 tail days) and $\alpha = 5\%$ (about 198)
- ▶ For (c): the eleven international banks (the six US banks of B1 and these five), each with its own regional system

□ Tasks

- (a) Compute VaR 1%, $\hat{b}_{1\%}$, ΔCoVaR 1% with a block-bootstrap interval (blocks of 20 days, 999 resamples) and ΔCoVaR 5%; rank the banks at each level and compute the Spearman correlation of the two rankings.
- (b) At each level, test equal ΔCoVaR for the 10 pairs on joint block-bootstrap draws with the Holm correction, and for all five banks jointly with a Wald test (χ_4^2), at 5%.
- (c) Compute the Spearman correlation of VaR 1% and ΔCoVaR 1% across the eleven banks, with a 95% block-bootstrap interval (blocks of 20 days, 999 resamples).

B2 [Proposed]: ΔCoVaR of European Banks and the VaR Ranking — Task (2/2)

□ Report

- ▶ A table of (a); both rankings and their Spearman correlation; the pair counts (raw, Holm), the Wald statistics, p -values and decisions; the Spearman correlation of (c) with its interval

□ Interpretation

- ▶ How many tail days does each level use?
- ▶ What does this do to the power of the tests?
- ▶ Would you name one European bank “most systemic” from these data?
- ▶ Is the VaR– ΔCoVaR correlation different from zero?
- ▶ Is it different from one?

□ Model: B1 [Solved]; notebook section B2

Team Project

- Choose a financial system not covered in class (e.g. Central and Eastern European banks, insurers, crypto exchanges)
- Estimate MES, ΔCoVaR and a connectedness table, each with block-bootstrap uncertainty; one historical and one reverse stress test
- Deliver a Quantlet folder (code, a Metainfo.txt description file, charts) and three slides: numbers, interpretation, limitations
- Log AI use in `AI_USE.md` and caught errors in `AI_ERRORS.md`; extension packs are optional

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. CoVaR variants and backtests	A2, B10	A1, B1
2. SRISK	B3 [Solved], A4, B4	A3; the lecture section on SRISK
3. Connectedness	A6, B6, C3	A5, B5, B9
4. Tail networks	B7 [Solved]	Lecture 18, Financial Risk Meter
5. Stress testing	A7 [Solved], A8, B8	Lecture 18, reverse stress test
6. Romanian banks	C1	B1, B9

- Each pack has its own notebook sections; do the exercises in the order of the pack
- Useful for the team project: C1 and C3 are possible project starts

A2 [Proposed]: The Girardi–Ergün CoVaR under the Normal Distribution — Task

- **Question:** How much larger is CoVaR when we condition on losses at least equal to VaR?
- **Data and assumptions**
 - ▶ Same bivariate Normal model and numbers as A1(c): $\sigma_{sys} = 1.5\%$, $\rho = 0.6$, $\alpha = 1\%$
 - ▶ Definition of [Girardi & Ergün \(2013\)](#): distress is $X_i \leq q_\alpha(X_i)$ (at or beyond the VaR), and $\text{CoVaR}_\alpha^\leq = c$ solves $P(X^{sys} \leq -c \mid X_i \leq q_\alpha(X_i)) = \alpha$
 - ▶ $\Phi_2(u, v; \rho)$: the standard bivariate Normal cumulative distribution function (cdf) with correlation ρ
- **Tasks**
 - (a) Write the conditional probability as a joint probability divided by α , standardise both coordinates and show that c solves $\Phi_2(z_\alpha, -c/\sigma_{sys}; \rho) = \alpha^2$.
 - (b) Prove that $\text{CoVaR}_\alpha^\leq \geq \text{CoVaR}_\alpha$ for $\rho > 0$, using that the law of X^{sys} given $X_i \leq q_\alpha$ is a mixture, over $x \leq q_\alpha$, of the laws of A1(a).
 - (c) Solve the equation of (a) numerically by bisection on c and compare the result with $\text{CoVaR}_{1\%}$ of A1.
- **Report**
 - ▶ c , the standardised threshold $-c/\sigma_{sys}$ and the gap to A1 in points and in percent
- **Interpretation**
 - ▶ The gap is small under the Normal distribution: why would it be much larger in bank data?
- **Model:** A1 [Solved]; notebook section A2

B10 [Proposed]: Backtesting the Girardi–Ergün CoVaR — Task (1/2)

- **Question:** Would a backtest detect a badly calibrated CoVaR?
- **Data and assumptions**
 - ▶ The eleven international banks of B1–B2 with their regional systems; daily returns and sample as in B1
 - ▶ Each day $t \geq 1\,000$, using all earlier days: $\hat{q}_t$ is the empirical 5% quantile of X_i ; $\text{CoVaR}_t^{\leq}$ is minus the empirical 5% quantile of X^{sys} on the past days with $X_i \leq \hat{q}_t$ (historical version of A2) [Girardi & Ergün \(2013\)](#)
 - ▶ Joint hit $I_t = 1\{X_t^{\text{sys}} \leq -\text{CoVaR}_t^{\leq}, X_{i,t} \leq \hat{q}_t\}$; under H_0 , $P(I_t = 1) = \alpha^2 = 0.25\%$
- **Tasks**
 - (a) Count the joint hits of each bank and test $H_0 : P(I_t = 1) = 0.25\%$ at 5% with the Kupiec test [Banulescu-Radu, Hurlin, Leymarie & Scaillet \(2021\)](#), likelihood-ratio statistic $LR = -2[\ln L(0.25\%) - \ln L(\hat{\pi})] \sim \chi_1^2$.
 - (b) Find the rejection region of the test in terms of the number of hits, and compute its power, assuming independent Bernoulli hits: the probability of rejecting when the true joint rate is 0.5%.

B10 [Proposed]: Backtesting the Girardi–Ergün CoVaR — Task (2/2)

▣ Report

- ▶ The number of forecast days; expected and observed hits; LR and p for JPMorgan; the range of hits and the smallest p over the eleven banks; the rejection region and the power

▣ Interpretation

- ▶ Why is α^2 the right null rate?
- ▶ What does a non-rejection tell you here?

▣ Model: B1 (tests), A2 (CoVaR $^{\leq}$); notebook section B10

B3 [Solved]: SRISK: LRMES by Simulation — Task (1/2)

- **Question:** How much would JPMorgan lose in a one-month market crisis, today and in March 2020?
- **Data and assumptions**
 - ▶ JPMorgan (adjusted close) and the S&P 500 (close), daily log returns in %, 5 January 2010 – 18 September 2026 [Brownlees & Engle \(2017\)](#)
 - ▶ Crisis as in [Brownlees & Engle \(2017\)](#): arithmetic market return below $C = -10\%$ over $h = 22$ days; LRMES is the expected bank loss over the same h days in a crisis
 - ▶ For (c): the six-month scenario ($h = 126$ days, $C = -40\%$), the shortcut $1 - \exp(-18 \text{ MES}^2)$ of Lecture 18 and the prudential ratio $k = 8\%$
- **Tasks**
 - (a) Estimate a zero-mean GJR–GARCH(1,1) for each return series and a DCC(1,1) for their correlation, by two-step QML.
 - (b) Simulate 200 000 paths of 22 days from resampled pairs of standardised innovations, keep the crisis paths and compute LRMES with its 90% range (5% and 95% quantiles of the loss); repeat with data up to 31 March 2020.
 - (c) Repeat (b) with $h = 126$, $C = -40\%$ (400 000 paths), compare with the shortcut, and compute $L^* = 1 + (1 - k)(1 - \text{LRMES})/k$ for both.

B3 [Solved]: SRISK: LRMES by Simulation — Task (2/2)

□ Report

- ▶ The GJR and DCC estimates; the share of crisis paths; LRMES with its range, today and on 31 March 2020; the six-month LRMES against the shortcut; both L^*

□ Interpretation

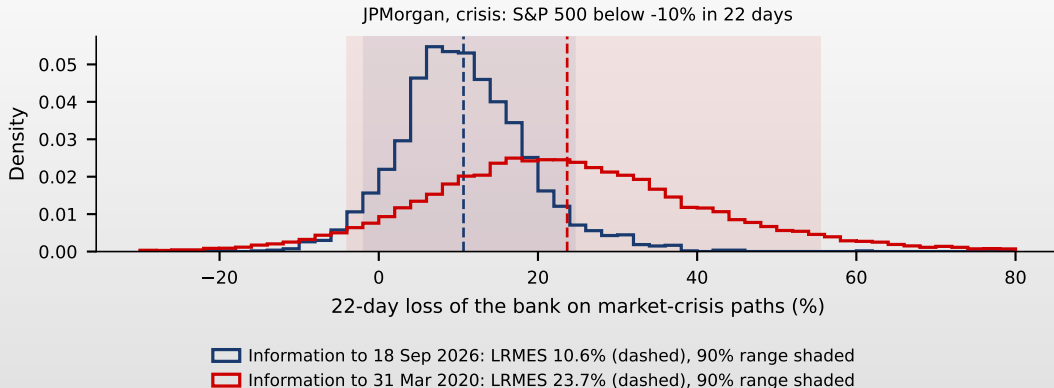
- ▶ Which assumption of the shortcut does the simulation drop?
- ▶ Why does LRMES change between March 2020 and today?
- ▶ What is missing to turn LRMES into SRISK in money?

□ Notebook: section B3

B3 [Solved]: Solution (1/3)

- (a) Variance: $\sigma_t^2 = \omega + (\alpha + \gamma 1\{r_{t-1} < 0\})r_{t-1}^2 + \beta\sigma_{t-1}^2$; JPMorgan: $\hat{\alpha} = 0.017$, $\hat{\gamma} = 0.120$, $\hat{\beta} = 0.891$, persistence $\hat{\alpha} + \hat{\gamma}/2 + \hat{\beta} = 0.967$.
- DCC on the standardised residuals: $Q_t = (1 - a - b)\bar{Q} + a\varepsilon_{t-1}\varepsilon_{t-1}^\top + bQ_{t-1}$; $\hat{a} = 0.052$, $\hat{b} = 0.928$; last correlation $\rho_T = 0.53$.
- (b) Steps: draw 22 days of pairs $(\xi_{i\tau}, \epsilon_{m\tau})$, with $\xi_{i\tau} = (\epsilon_{i\tau} - \rho_\tau \epsilon_{m\tau}) / \sqrt{1 - \rho_\tau^2}$; rebuild $\epsilon_i = \rho \epsilon_{m\tau} + \sqrt{1 - \rho^2} \xi_{i\tau}$ with the simulated ρ ; run the GJR and DCC recursions from today's state; keep the paths with market return below -10% ; average the bank's arithmetic loss over them.
- Result: 1.3% of the paths are crises (about 2 687 of 200 000); LRMES = 10.6%, 90% range $[-2, 25]\%$.
- With data up to 31 March 2020 the same procedure gives LRMES = 23.7%.

B3 [Solved]: Solution (2/3)



B3 [Solved]: Solution (3/3)

- Simulated 22-day loss of JPMorgan on the paths where the S&P 500 falls below -10% ; dashed: LRMES; shaded: 90% range
- (c) Shortcut: JPMorgan's $\text{MES}^{2\%} = 3.61\%$, $18 \cdot 0.0361 = 0.650$, so $\text{LRMES} = 1 - e^{-0.650} = 48\%$.
- Simulation for $h = 126$, $C = -40\%$: $\text{LRMES} = 33\%$.
- $L^* = 1 + 0.92(1 - 0.335)/0.08 = 8.7$ from the simulation against $1 + 0.92(1 - 0.478)/0.08 = 7.0$ from the shortcut.
- Interpretation: the simulation drops the constant beta and the log-return scaling behind the 18 of the shortcut (see A4); it starts from today's volatility and correlation and keeps the joint tail of the innovations.
- On 31 March 2020 volatility and correlation were at crisis levels, so a further -10% month would have caused a larger bank loss; today the starting state is calm.
- SRISK in money also needs the book value of debt D and the market value of equity W .

A4 [Proposed]: Where the 18 in the LRMES Shortcut Comes From — Task (1/2)

□ **Question:** Where does the constant 18 in the LRMES shortcut come from?

□ **Data and assumptions**

- ▶ Crisis scenario [Acharya, Engle & Richardson \(2012\)](#): the market falls by exactly $d = 40\%$ over six months, a deterministic boundary scenario ($R_m = -d$) that motivates the shortcut; the simulated LRMES of B3 instead averages over all paths with $R_m < C$
- ▶ The bank's six-month log return is β times the market's, with constant β ; $\text{MES}^{2\%} = \beta \text{ES}_m^{2\%}$, where $\text{ES}_m^{2\%}$ is the average market loss on days with a return below -2%
- ▶ Data for (c): S&P 500 daily log returns (close), 5 January 2010 – 18 September 2026; and a Normal market with $\sigma_m = 1\%$

□ **Tasks**

- (a) Show that in this scenario the bank loses $\text{LRMES} = 1 - (1 - d)^\beta$.
- (b) Substitute $\beta = \text{MES}_m^{2\%} / \text{ES}_m^{2\%}$ to show $\text{LRMES} = 1 - \exp(-\kappa \text{MES}_m^{2\%})$ with $\kappa = -\ln(1 - d) / \text{ES}_m^{2\%}$.
- (c) Compute $\text{ES}_m^{2\%}$ and κ for the S&P 500 and for the Normal market ($\text{ES}_m^{2\%} = \sigma_m \varphi(2) / \Phi(-2)$), and the $\text{ES}_m^{2\%}$ that gives $\kappa = 18$.

A4 [Proposed]: Where the 18 in the LRMES Shortcut Comes From — Task (2/2)

□ Report

- ▶ The number of S&P 500 days below -2% , both values of $ES_m^{2\%}$ and of κ , and the $ES_m^{2\%}$ implied by 18

□ Interpretation

- ▶ What goes wrong in the shortcut if β rises in crises?

□ Model: A3 [Solved]; notebook section A4

B4 [Proposed]: SRISK of a European and a Romanian Bank — Task

- **Question:** How exposed are Deutsche Bank and Banca Transilvania to a crisis of their own market?
- **Data and assumptions**
 - ▶ Deutsche Bank with the Euro Stoxx 50 as its market; Banca Transilvania with BET; sample of B3
 - ▶ Models and crisis as in B3: GJR-GARCH(1,1), DCC(1,1), 200 000 paths of 22 days, $C = -10\%$
 - ▶ Prudential ratio $k = 5.5\%$ Engle, Jondeau & Rockinger (2015), so $L^* = 1 + 0.945(1 - \text{LRMES})/0.055$
- **Tasks**
 - (a) Compute LRMES with its 90% range for both banks with data up to 18 September 2026 and up to 31 March 2020, and the share of crisis paths of each market.
 - (b) Compute L^* from the LRMES and from both ends of its 90% range.
- **Report**
 - ▶ LRMES with range (both dates), crisis shares, L^* with its range, for both banks
- **Interpretation**
 - ▶ Which bank is more exposed to a crisis of its own market?
 - ▶ How wide is the range of L^* ?
 - ▶ What does this range mean for a supervisor?
- **Model:** B3 [Solved]; notebook section B4

A6 [Proposed]: Connectedness of a Bivariate VAR — Task

□ **Question:** Who transmits and who receives in a two-bank VAR?

□ **Data and assumptions**

- ▶ VAR(1) for two banks with $A = \begin{pmatrix} 0.5 & 0 \\ 0.4 & 0.5 \end{pmatrix}$ and $\Sigma = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix}$
- ▶ $A_{21} = 0.4$, $A_{12} = 0$: bank 1 drives bank 2 with a one-day lag, not the reverse
- ▶ Generalized decomposition of A5, rows normalised to one; $C = \frac{100}{2}(\tilde{\theta}_{12} + \tilde{\theta}_{21})$; net position $T_i - F_i$ (to others minus from others)

□ **Tasks**

- (a) Compute the generalized table (raw and normalised) and C at $H = 1$.
- (b) Compute $A\Sigma$, $\text{diag}(A\Sigma A^\top)$, the raw and normalised tables and C at $H = 2$.
- (c) Compute the net position of each bank at $H = 2$ and identify the net transmitter.

□ **Report**

- ▶ Both tables, C at both horizons and the two net positions

□ **Interpretation**

- ▶ Why is the table symmetric at $H = 1$ although bank 1 drives bank 2?
- ▶ Why does C rise with H ?

□ **Model:** A5 [Solved]; notebook section A6

B6 [Proposed]: Did Connectedness Change after 2020? — Task (1/2)

- **Question:** Did bank connectedness rise after 2020?
- **Data and assumptions**
 - ▶ Rolling total connectedness of Lecture 18: log Parkinson volatilities of the 13 banks (B5), windows of 250 days moved by 5 days, VAR(2), $H = 10$; 744 windows
 - ▶ Break date: 1 January 2020; for (b) the daily series of B5 in 2010–2019 (2 373 days) and in 2020–2026 (1 593 days)
- **Tasks**
 - (a) Regress the rolling index on a constant and $D_t = 1\{t \geq 2020\}$ by OLS; test H_0 : slope = 0 at 5% with ordinary and with Newey–West (50 lags) standard errors [Newey & West \(1987\)](#).
 - (b) Estimate a VAR(1) separately on 2010–2019 and on 2020–2026 and compute C ($H = 10$) of each, with 95% intervals from a residual block bootstrap (blocks of 20 days, 999 draws) in each period.
 - (c) Build a 95% interval for the difference $C_{2020-26} - C_{2010-19}$ from the differences of independent draws in the two periods, and decide at 5%.

B6 [Proposed]: Did Connectedness Change after 2020? — Task (2/2)

□ Report

- ▶ The two means, the slope, both t statistics and decisions; C for both periods with intervals; the difference with its interval

□ Interpretation

- ▶ Why are consecutive values of the rolling index nearly copies of each other?
- ▶ Why is (b)–(c) a cleaner test than (a)?
- ▶ What does it still assume about the break date?

□ Model: B5, B9 [Solved]; notebook section B6

C3 [Proposed]: Quantile Connectedness of Bank Losses — Task (1/2)

- **Question:** Do large bank losses spread more than small ones?
- **Data and assumptions**
 - ▶ Case study of Lecture 18: [Ando, Greenwood-Nimmo & Shin \(2022\)](#), spillover index of a quantile factor VAR (QFVAR) at quantile τ and $\text{RTD} = S_{0.95}^{(5)} - S_{0.05}^{(5)}$ (Sections 2 and 4.3)
 - ▶ Daily adjusted closes of the 13 banks of B5 (course dataset), 2010–2026, aligned on common days before computing returns; $y_{it} = -100 \ln(P_{it}/P_{i,t-1})$, the daily loss in %, so that the right tail is adverse, as for the CDS changes of the paper
 - ▶ Generalized FEVD of Eq. (17), rows normalised, $h = 5$ days
- **Tasks**
 - (a) Build the loss panel, choose the VAR order by BIC on the full sample and report it together with the number of factors selected by Eq. (8) at the mean and at each τ of (b) (the number is quantile-specific).
 - (b) Estimate the factor VAR at the mean and at $\tau \in \{0.05, 0.10, 0.50, 0.90, 0.95\}$ on the full sample, and compute the spillover index $S_{\tau}^{(5)}$ of Eq. (20) at each.
 - (c) Repeat (b) without factors, as a plain quantile VAR, and compare the indices; interpret their difference as sensitivity to common-factor adjustment, not as an identified common-shock contribution.
 - (d) On rolling windows of 250 days moved by one day, with the number of factors fixed at its full-sample value, compute $S^{(5)}$ at the mean, 0.05, 0.5 and 0.95, and RTD; this is descriptive: the full-sample choices use future data.

C3 [Proposed]: Quantile Connectedness of Bank Losses — Task (2/2)

▣ Report

- ▶ Indices of (b)–(c) beside Figure 4 of the paper; correlations of the rolling indices of (d) beside Table 3, with block-bootstrap intervals; a chart of RTD with March 2020 and March 2023 marked; the data limitation (equity losses, 13 banks, different closing times)

▣ Interpretation

- ▶ Do bank losses spread more in both tails than in the centre, as in the credit risk network?
- ▶ Does RTD rise before the stress of 2020 and 2023 or only with it?

▣ Model: B5, B9, Lecture 18 (case study); notebook section C3

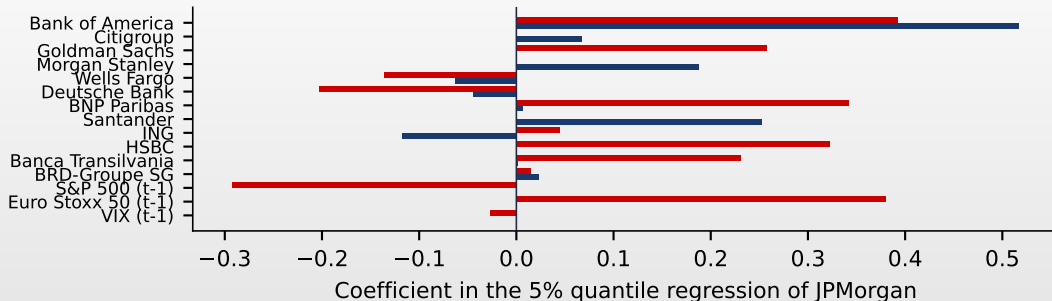
B7 [Solved]: The FRM Penalty in One Window — Task

- **Question:** Which banks carry JPMorgan's left tail, in a calm and in a crisis window?
- **Data and assumptions**
 - ▶ Daily returns (fractions) of the 13 banks and of the S&P 500, Euro Stoxx 50 and VIX lagged by one day
 - ▶ FRM regression [Mihoci et al. \(2020\)](#): JPMorgan on the other 12 banks and the three lagged indices (15 regressors), quantile level $\tau = 5\%$, L_1 penalty λ , windows of 63 days
 - ▶ Grid of 20 values of λ log-spaced from 10^{-6} to $10^{-1.5}$; $GACV(\lambda) = \sum_t \rho_\tau(\hat{u}_t)/(n - \text{df})$, df = number of non-zero coefficients
- **Tasks**
 - (a) In the calm window ending 28 June 2019, fit the penalised quantile regression for each λ on the grid, choose the λ that minimises GACV and count the active coefficients.
 - (b) Repeat (a) in the crisis window ending 31 March 2020, and list the four largest coefficients in absolute value in each window.
- **Report**
 - ▶ $\hat{\lambda}$, the number of active coefficients and the four largest coefficients in each window
- **Interpretation**
 - ▶ Which banks carry JPMorgan's tail in each window?
 - ▶ Why is the selected λ so small?
- **Notebook:** section B7

B7 [Solved]: Solution (1/3)

- Steps: (1) take the 63 days of the window; (2) for each λ solve the penalised quantile regression; (3) compute the residuals, df and GACV; (4) keep the λ with the smallest GACV.
- (a) Calm window (2019-03-21 to 28 June 2019): $\hat{\lambda} = 4.5 \times 10^{-5}$, 11 of 15 coefficients active.
- Largest: Bank of America (0.52), Santander (0.25), Morgan Stanley (0.19), ING (-0.12).
- (b) Crisis window (2019-12-23 to 31 March 2020): $\hat{\lambda} = 7.8 \times 10^{-5}$, 12 of 15 active.
- Largest: Bank of America (0.39), Euro Stoxx 50 (t-1) (0.38), BNP Paribas (0.34), HSBC (0.32).

B7 [Solved]: Solution (2/3)



■ Calm window ending 28 Jun 2019: $\lambda = 4.5e-05$, 11 of 15 active

■ COVID-19 window ending 31 Mar 2020: $\lambda = 7.8e-05$, 12 of 15 active

B7 [Solved]: Solution (3/3)

- The 15 coefficients at the GACV-selected penalty in the two windows
- Interpretation: Bank of America leads in both windows; in the crisis the lagged Euro Stoxx 50 and two European banks join the largest coefficients: the tail of JPMorgan becomes international.
- $\hat{\lambda}$ rises from calm to crisis, the direction the FRM reads as higher tail risk.
- λ is small because 15 regressors for 63 days make the unpenalised fit feasible: GACV prefers little shrinkage.
- The FRM is designed for many more firms than days, where the penalty is indispensable [Toma et al. \(2026\)](#).

 [MFM_ch18_seminar](#)

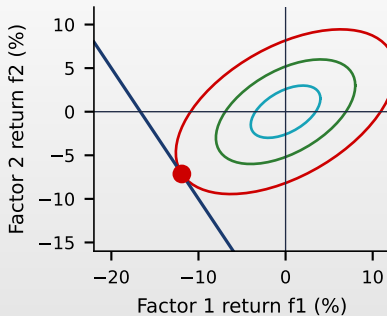
A7 [Solved]: Reverse Stress Test: the Lagrangian — Task

- **Question:** What is the most plausible scenario behind a given portfolio loss?
- **Data and assumptions**
 - ▶ Two risk factors $f = (f_1, f_2)^\top \sim N(0, \Sigma)$ (four-week returns, %); portfolio loss $L = -b^\top f$ with exposures b
 - ▶ Reverse stress test: the most plausible scenario (highest density) that produces a given loss ℓ^*
 - ▶ Numbers for (c): $b = (1.2, 0.8)$, $\Sigma = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix}$ ($\%^2$), $\ell^* = 20\%$
- **Tasks**
 - (a) Minimise $\frac{1}{2}f^\top \Sigma^{-1}f$ subject to $-b^\top f = \ell^*$ with a Lagrangian and derive f^* .
 - (b) Show that the Mahalanobis distance of f^* is $d^* = \ell^* / \sqrt{b^\top \Sigma b}$ and that $P(L \geq \ell^*) = \Phi(-d^*)$.
 - (c) Compute Σb , $b^\top \Sigma b$, f^* , d^* and $P(L \geq \ell^*)$, and check that f^* produces the target loss.
- **Report**
 - ▶ Every intermediate vector and scalar, f^* , d^* and the probability
- **Interpretation**
 - ▶ Which factor carries the larger fall in the scenario?
 - ▶ How should a risk committee read a probability of this size?
- **Notebook:** section A7

A7 [Solved]: Solution (1/3)

- (a) Maximum density is minimum $\frac{1}{2}f^\top \Sigma^{-1}f$; Lagrangian $\mathcal{L} = \frac{1}{2}f^\top \Sigma^{-1}f + \lambda(b^\top f + \ell^*)$.
- First-order condition: $\Sigma^{-1}f + \lambda b = 0$, so $f = -\lambda \Sigma b$.
- Constraint: $-b^\top f = \lambda b^\top \Sigma b = \ell^*$, so $\lambda = \ell^* / (b^\top \Sigma b)$ and $f^* = -\ell^* \Sigma b / (b^\top \Sigma b)$.
- (b) $f^{*\top} \Sigma^{-1} f^* = \lambda^2 b^\top \Sigma b = \ell^{*2} / (b^\top \Sigma b)$, so $d^* = \ell^* / \sqrt{b^\top \Sigma b}$.
- $L = -b^\top f \sim N(0, b^\top \Sigma b)$, so $P(L \geq \ell^*) = 1 - \Phi(\ell^* / \sqrt{b^\top \Sigma b}) = \Phi(-d^*)$.

A7 [Solved]: Solution (2/3)



— Mahalanobis distance 1 — Mahalanobis distance 3.15 — Loss line $-b'f = 20\%$ ● $f^* = (-11.9, -7.1)$
— Mahalanobis distance 2

A7 [Solved]: Solution (3/3)

- Ellipses: points of equal density of f ; the most plausible scenario is where the loss line touches the smallest ellipse
- (c) $\Sigma b = (16 \cdot 1.2 + 6 \cdot 0.8, 6 \cdot 1.2 + 9 \cdot 0.8) = (24, 14.4)$.
- $b^\top \Sigma b = 1.2 \cdot 24 + 0.8 \cdot 14.4 = 40.32$, $\sqrt{40.32} = 6.35$;
 $\lambda = 20/40.32 = 0.496$.
- $f^* = -0.496 \cdot (24, 14.4) = (-11.9\%, -7.1\%)$; check:
 $1.2 \cdot 11.90 + 0.8 \cdot 7.14 = 20\%$.
- $d^* = 20/6.35 = 3.15$ and
 $P(L \geq 20\%) = \Phi(-3.15) = 0.082\%$.
- Interpretation: factor 1 falls by 11.90% (2.98 of its standard deviations), factor 2 by 7.14% (2.38): the scenario leans on the factor with the larger exposure and variance.
- Under the Normal model such a four-week loss happens about once in 94 years; A8 shows how fragile this number is.

A8 [Proposed]: Reverse Stress Test under a Student- t Distribution — Task

- **Question:** Does the reverse stress scenario survive heavy tails?
- **Data and assumptions**
 - ▶ Same b and Σ as A7; targets $\ell^* = 20\%$ and $\ell^* = 30\%$
 - ▶ f is multivariate Student- t with $\nu = 4$ degrees of freedom and covariance Σ , so its scale matrix is $\Sigma(\nu - 2)/\nu$; T_ν is the Student- t cumulative distribution function (cdf)
- **Tasks**
 - (a) Show that the most plausible scenario f^* is the same as in A7, using that the density depends on f only through $f^\top \Sigma^{-1} f$ and decreases in it.
 - (b) Show that $L = -b^\top f$ is a univariate t_ν with scale $\sqrt{b^\top \Sigma b (\nu - 2)/\nu}$, so that $P(L \geq \ell^*) = 1 - T_\nu(d^* \sqrt{\nu/(\nu - 2)})$.
 - (c) Compute the probability for both targets under the Normal and the t_4 models, and the ratio of the two.
- **Report**
 - ▶ d^* , the t argument, the two probabilities and their ratio for each target
- **Interpretation**
 - ▶ What is robust in a reverse stress test?
 - ▶ What is not robust?
- **Model:** A7 [Solved]; notebook section A8

B8 [Proposed]: Stress Test of a Romanian Bank Portfolio — Task (1/2)

- **Question:** What would a severe but plausible shock cost a Romanian bank portfolio?
- **Data and assumptions**
 - ▶ Portfolio of 1 000 000 RON, half in Banca Transilvania and half in BRD, rebalanced daily; daily log returns, days with no trades removed
 - ▶ Historical windows: the bank-tax shock (10 December 2018 – 31 January 2019) and COVID-19 (15 February – 30 April 2020)
 - ▶ Hypothetical scenario: weekly log returns of -25% for the Euro Stoxx 50 and -20% for BET; weekly (Friday) log returns 2010–2026 of both banks, the Euro Stoxx 50 and BET
- **Tasks**
 - (a) The daily log return of the rebalanced portfolio is $r_p = 100 \ln\left(\frac{1}{2}e^{r_{TLV}/100} + \frac{1}{2}e^{r_{BRD}/100}\right)$, not the average of the two log returns; in each window find the worst sum S of 10 consecutive r_p and convert it into a loss $1 - e^{S/100}$, in % and in RON.
 - (b) Regress each bank's weekly return on the Euro Stoxx 50 (β_1) and BET (β_2) weekly returns by OLS with a constant; predict each bank's factor-induced log return $\hat{r}_k = \beta_{1k}(-25) + \beta_{2k}(-20)$ (intercept and residual set to zero), convert it to a simple return $e^{\hat{r}_k/100} - 1$ and apply the 50/50 weights held unchanged over the scenario week; also report the average betas and R^2 .

B8 [Proposed]: Stress Test of a Romanian Bank Portfolio — Task (2/2)

- ▣ **Report**

- ▶ Both historical losses with their dates and RON amounts; the betas, R^2 and the hypothetical loss in % and RON

- ▣ **Interpretation**

- ▶ Which scenario hurts more: a domestic political shock or a global pandemic?
- ▶ Why is the Euro Stoxx 50 beta close to zero once BET is in the model?

- ▣ **Model:** A7 [Solved]; notebook section B8

C1 [Proposed]: How Exposed Are Romanian Banks to European Banking Stress? — Task (1/2)

- **Question:** Are Romanian banks exposed mainly to European banking stress or to domestic shocks?
- **Data and assumptions**
 - ▶ Banca Transilvania and BRD; European system: equal-weighted portfolio of Deutsche Bank, BNP Paribas, Santander, ING and HSBC; BET
 - ▶ Sample: 5 January 2010 – 18 September 2026, 3 960 trading days common to all 13 banks and the four indices (prices aligned on common days first, then daily log returns in %)
 - ▶ Exposure CoVaR: quantile regression of the Romanian bank on the European system, so ΔCoVaR is the extra loss of the Romanian bank when the European system is in distress (the reverse direction of B1)
- **Tasks**
 - (a) Estimate the exposure ΔCoVaR of each Romanian bank at 5% and at 1% with block-bootstrap intervals (blocks of 20 days, 999 resamples), and the same measure with BET in place of the European system.
 - (b) Split the sample into 2010–2017 and 2018–2026, estimate ΔCoVaR 5% in each, and test the change at 5% with $z = (\hat{\Delta}_2 - \hat{\Delta}_1) / \sqrt{\text{se}_1^2 + \text{se}_2^2}$ from independent block-bootstrap draws (or a dominance test [Castro & Ferrari \(2014\)](#)), not by overlapping intervals.
 - (c) With rolling Diebold–Yilmaz tables (log Parkinson volatilities of the 13 banks, windows of 250 days moved by 5 days, VAR(2), $H = 10$), compute the share of the Romanian banks' forecast error variance coming from European and from US banks.

C1 [Proposed]: How Exposed Are Romanian Banks to European Banking Stress? — Task (2/2)

□ **Report**

- ▶ Deliverable: a Quantlet folder (code, a Metainfo.txt description file, charts) and three slides: the numbers of (a)–(c) with intervals and tests, the interpretation, the limitations

□ **Interpretation**

- ▶ Are Romanian banks exposed mainly to European banking stress or to domestic shocks?
- ▶ Has the exposure grown?

□ **Notebook: section C1**

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